EVALUATING THE UTILITY OF SYNTHETIC T1C BRAIN MRI SCANS GENERATED BY GAN AND DIFFUSION MODELS

JULIAN OLBINSKI

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STUDENT NUMBER

2084796

COMMITTEE

dr. Sharon Ong

Msc. Chiara Manna

LOCATION

Tilburg University

School of Humanities and Digital Sciences

Department of Cognitive Science &

Artificial Intelligence

Tilburg, The Netherlands

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Abstract

Glioma is the most common malignant brain tumor and carries a poor prognosis, making non-invasive assessment with MRI critical in patient care. Specifically, T1-weighted post-contrast (T1c) MRI scans are essential for accurate delineation of glioma subregions and tumor grading. However, they are frequently absent due to limited scanner availability, cost, and contrast agent contraindications. This negatively impacts patient care, and also the use and training of segmentation models, which rely on complete data to accurately delineate tumor regions and support treatment planning. Generative models have recently been applied to MRI synthesis to alleviate modality scarcity, but T1c synthesis remains underexplored due to heterogeneous contrast enhancement patterns that are challenging to learn. This thesis investigates whether state-of-the-art models can synthesize T1c scans that are both realistic and capable of substituting for missing T1c in a segmentation task. One unconditional and three conditional models were trained and assessed on the BraTS-2023 Adult Glioma dataset, with conditional models evaluated for generalization on the external UCSF-PDGM dataset. Two conditional models successfully synthesized high-fidelity T1c, outperforming missing T1c conditions when used in segmentation, while the third conditional model failed T1c synthesis. Segmentation performance degraded most for enhancing tumor regions and small tumors when using synthetic scans, indicating disparate clinical impact for patients who rely on assessment of tumors with such features. Finally, the strong performance of the unconditional model suggests the T1c distribution is learnable. These results support T1c synthesis as partially clinically useful, with enhancing tumor learning identified as a remaining challenge.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

The primary data used in this publication were obtained as part of the Brain Tumor Segmentation (BraTS) Challenge project through Synapse ID: syn51156910 (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2015). Data were collected and compiled with consent from adult glioma patients at various hospitals. This study also used the images and annotations subset of the UCSF-PDGM dataset (Calabrese et al., 2023), which is hosted by the Cancer Imaging Archive (Clark et al., 2013) and licensed under CC by 4.0. The use of BraTS 2023 Adult Glioma data and the UCSF-PDGM data in this project is consistent with the Attribution-NonCommercial 4.0 International deed, as it is used strictly for research and not intended for commercial use. All data were previously anonymized. Both datasets primarily represent western demographics. No further image data was collected or acquired. Aside from the use of BraTS and UCSF-PDGM data for comparison in several figures, as well as images generated using generative models cited below, all images belong to the author.

All model codes were sourced via open source repositories provided by the authors (Conte et al., 2021; Isensee et al., 2021; Karras et al., 2020; Mirowski & Fabijańska, 2026; Özbey et al., 2023). Code was adapted as necessary to suit the needs of the experiments conducted in this study. The repositories and corresponding original code are provided in the footnotes.[1] [2] [3] [4] [5]. Data preprocessing was conducted according to the individual needs of the generative models. The code used for this thesis is available upon request.

The main Python libraries used in this study were PyTorch/torchvision (Paszke et al., 2019), MONAI (Cardoso et al., 2022), Scikit-learn (Pedregosa et al., 2011), Scikit-image (Walt et al., 2014), matplotlib (Hunter, 2007), TensorFlow (Abadi et al., 2016), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), and Pandas (McKinney, 2010). Full libraries and associated versions are available in Appendix 6.10.

Microsoft Word spellcheck was used to correct spelling errors. The thesis was adapted to and presented in Overleaf. References were managed with Zotero. Draw.io was used in creating figures. Claude.ai (Anthropic, 2024) was used to aid in the construction of code, presentation of figures and tables, as well as improve writing. AI-generated text was revised and integrated by the author.

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1 https://github.com/giemmecci/pix2pixRAD
2 https://github.com/NVlabs/stylegan2-ada-pytorch
3 https://github.com/icon-lab/SynDiff
4 https://github.com/Miretpl/synthetic-brain-mri-project
5 https://github.com/MIC-DKFZ/nnUNet

2 PROBLEM STATEMENT AND RELEVANCE

2.1 *Clinical Burden of Brain Tumors*

Brain tumors are often highly aggressive, with an average 5-year survival rate of roughly 35% for malignant tumors (Ostrom et al., 2022). Of these, gliomas are the most common, representing roughly 81% of cases (Ostrom et al., 2014, 2022). Gliomas are classified by the WHO according to severity, with low-grade (LGG) corresponding to grades I–II, and high-grade (HGG) to grades III–IV (Louis et al., 2021). Glioblastomas, a subtype of HGG, are the most malignant and common subtype of glioma, representing 40-50% of total cases and carrying a median overall survival of 14-16 months and a 5-year survival rate of below 10% (Aleid et al., 2024; Ostrom et al., 2022).

Even with interventions like surgery and chemotherapy, survival rates remain low (Ostrom et al., 2022). However, outcomes are significantly improved with early diagnosis and treatment (Thenuwara et al., 2023). Magnetic resonance imaging (MRI) has emerged as an indispensable tool clinicians use to diagnose, assess, and segment brain tumors non-invasively (Falk Delgado, 2025; Wei & Wei, 2021); and segmentation models trained using MRI have helped clinicians by improving tumor segmentation accuracy and reducing time and effort (Isensee et al., 2021; Laukamp et al., 2021).

2.2 *T1c Scan Importance and Missingness*

Brain MRI scans are multi-modal, with modalities often examined together to allow for more accurate clinical assessment (Kim & Park, 2024). Among these, T1-weighted post-contrast (T1c) scans are widely used in glioma imaging (Hirschler et al., 2023). By using gadolinium as a contrasting agent to highlight sites of blood-brain barrier disruption, frequently seen in HGG, T1c enables radiologists to properly delineate glioma grade and anatomy (Jayachandran Preetha et al., 2021; Sarkaria et al., 2018; Warntjes et al., 2018).

Despite their importance, T1c scans are often unavailable due to limited scanner availability, differing protocols, scan time, and cost; and are therefore frequently absent from both clinical and research datasets (Kim & Park, 2024; Nafi et al., 2024; Zhou et al., 2022). Unfortunately, conventional data augmentation methods cannot adequately compensate for sparse unique data, and missing T1c cannot be directly replaced by non-contrast modalities at inference time, motivating the use of generative models to synthesize missing modalities (Islam et al., 2026; Li et al., 2024; Nafi et al., 2024; Raut et al., 2024).

2.3 *Problem Statement*

This study addresses the problem that T1c scans, critical for glioma assessment, are often missing, adversely affecting patient care and clinical research. The central goal is to investigate whether state-of-the-art (SOTA) generative models can synthesize T1c that are realistic and capable of substituting for real T1c in downstream glioma segmentation.

2.4 *Generative Models as a Solution*

Brain MRI synthesis is dominated by generative adversarial networks (GAN) and diffusion models, which address scarcity by learning the underlying image distribution to synthesize realistic images (Kazerouni et al., 2023; Nafi et al., 2024; Tavse et al., 2022). GANs consist of two competing networks: a generator, which produces synthetic images; and a discriminator, which attempts to distinguish real from synthetic images. These are trained jointly until the generator can generate images that fool the discriminator (Goodfellow et al., 2014). However, they may be sensitive to training instability (e.g., mode collapse) and may struggle to synthesize diverse images (Saad et al., 2022; Skandarani et al., 2023).

Diffusion models instead learn to generate images by reversing a gradual noising process (Ho et al., 2020). While they have been found to produce more diverse, higher-fidelity brain MRI images than GANs (Dorjsembe et al., 2024; Müller-Franzes et al., 2023); they do so at a greater computational cost (Kazerouni et al., 2023; Usman Akbar et al., 2024).

Both can be broadly categorized into two types. Conditional models use existing scans as input, using them to map onto a different target — e.g., using T1c to generate a non-contrast scan. This makes them suitable for generating substitute scans when a scan is missing (Raut et al., 2024). Unconditional models instead learn the underlying image distribution of a single target and generate novel images through random sampling, making them suited for dataset augmentation (Motamed et al., 2021; Tariq et al., 2023). This study investigates these model families and types in the context of T1c synthesis.

2.5 *Societal Motivation*

The societal motivation behind this study is to contribute to positive treatment outcomes in glioma patients. For the patient, absent T1c reduces the ability to characterize enhancement patterns common in HGGs and critical in treatment planning (Hirschler et al., 2023; Unkelbach et al., 2014). Additionally, gadolinium has several contraindications, including severe complications for those with kidney disease, fetal development risks in pregnant women, and long-term risks of gadolinium accumulation (Bird et al., 2019; Jayachandran Preetha et al., 2021; Schieda et al., 2018). The ability to generate high-fidelity substitute T1c may overcome resource barriers and help vulnerable patients avoid undue risks.

2.6 *Scientific Motivation*

The scientific motivation of this study is to contribute to the improvement of patient-level T1c supplementation at inference time. Recently, the BraSyn challenge has explicitly targeted this scenario, aiming to substitute missing MRIs with synthetic scans so that segmentation models can run on complete multi-modal inputs, rather than suffer degraded performance due to missing scans (Jain et al., 2026; Li et al., 2024). In this context, synthetic T1c could stand in for missing T1c during segmentation, potentially improving care for patients who don't have access to it (Raut et al., 2024).

However, T1c synthesis is underrepresented in brain MRI synthesis literature. Models developed for other scans often struggle to generalize and are often not evaluated on T1c synthesis (Dayarathna et al., 2024; Ibrahim et al., 2025; Meng et al., 2024). Furthermore, only few recent studies explicitly target T1c synthesis (Dong et al., 2026; Eidex et al., 2025). Also, synthetic brain MRI is often not evaluated on downstream clinical utility (Mukherkjee et al., 2022; Wu et al., 2024). Since segmentation models rely on large, complete datasets to generalize well (Kim & Park, 2024; Raut et al., 2024), T1c missingness can degrade their performance. Therefore, it remains unknown if SOTA generative models can synthesize high-fidelity T1c which can supplement missing T1c in downstream segmentation.

2.7 *Research Strategy*

The research aims of this study serve to address the societal and scientific motivations presented above by systematically evaluating synthetic T1c for both image fidelity and downstream segmentation, emphasizing patient-level supplementation at inference time.

RQ1: Can SOTA generative models generate T1c images that are comparable in quality to ground truth T1c?

This RQ seeks to evaluate if SOTA models — unconditional and conditional — can synthesize T1c. Both model type and conditioning strategy were evaluated, the latter treated as a form of input feature selection, in order to assess impact on T1c synthesis. Following training, synthetic image fidelity was measured across the following dimensions: distribution quality, pixelwise and structural similarity, and perceptual fidelity.

SQ1: Can an unconditional model learn the T1c distribution? The unconditional model serves to examine to what degree the T1c distribution is learnable without conditioning inputs. This would establish whether failure to learn conditional T1c generation is due to T1c itself, or if it is a function of model architecture and/or conditioning strategy. It was evaluated on distributional quality, since its outputs cannot be directly evaluated against ground-truth scans.

SQ2: Can conditional models generalize to an external dataset? Generalizability to unseen data is a key limitation of image synthesis studies. Testing against an external dataset helps determine the extent to which models can adapt to other image distributions. Conditional models trained on the primary dataset generated images after conditioning on an external dataset. Fidelity assessment was identical to RQ1.

RQ2: How well can synthetic conditional T1c images substitute for missing T1c in downstream segmentation?

Synthetic T1c replacement in a segmentation task more directly assesses potential clinical use than image fidelity metrics alone. To evaluate how well synthetic T1c can substitute for missing T1c, a segmentation model was trained exclusively on complete, real multi-modal glioma MRI data — approximating a real-world clinical segmentation scenario — and was applied to five distinct conditions, all of which used complete non-contrast MRIs but with T1c data that was systematically varied. Three reflected realistic T1c data availability: 0% missing T1c; 50% missing T1c; and 100% missing T1c. Two conditions reflected substitution, where missing T1c was replaced with synthetic scans at inference time: 50% real, 50% synthetic; and 100% synthetic T1c. Scans generated from each conditional model underwent segmentation in these two substitution conditions.

SQ3: How does synthetic T1c segmentation performance vary by tumor size/region? Segmentation performance is known to vary by tumor size, which affects the amount of visible enhancing tissue, and subregions, which reflect different tissue types. SQ2 examines the disparate impact across these dimensions, identifying whether performance degrades with respect to tumor size and subregion with synthetic T1c supplementation. Systematic underperformance for specific phenotypes — like small tumors or enhancing tumor regions — would disproportionately affect patients presenting with them, who would rely on their accurate segmentation for proper assessment. RQ2 segmentation scores were therefore stratified by tumor size and subregion.

SQ4: Does synthetic T1c segmentation performance generalize to an external dataset? Following the logic presented in SQ2, SQ4 seeks to evaluate how well SOTA models can generalize to a segmentation task with a novel dataset. This will employ identical strategies to RQ2 on an external dataset.

2.8 *Findings*

This study found that synthetic T1c scans can partially compensate for missing T1c in glioma segmentation, but performance varied based on generative architecture, conditioning strategy, tumor region, tumor size, and dataset used. Using synthetic T1c in place of missing T1c improved segmentation compared to missing T1c conditions – but it did not achieve the same performance as using only real T1c.

3 LITERATURE REVIEW

3.1 *T1c: Clinical Importance and Synthesis Challenges*

Gadolinium enhancement patterns are specific to T1c, making them critical for effective glioma assessment — i.e., differentiating HGG from LGG, and clarifying the extent of tumor subregions (Jayachandran Preetha et al., 2021; Osman & Tamam, 2023; Tabassum et al., 2024). However, this advantage is what makes them challenging to synthesize. Contrasts are highly heterogeneous, varying based on tumor type and individual anatomical differences, and they have no equivalent in non-contrast scans from which to map (Dong et al., 2026; Eidex et al., 2025; Tabassum et al., 2024). Additionally, while generative models may cohere to the overall anatomical structure, they may struggle to learn more subtle tumor anatomy (Dong et al., 2026; Frid-Adar et al., 2018).

3.2 *State-of-the-Art Brain MRI Synthesis*

Given the need to address T1c missingness and the challenges associated with T1c synthesis, this study evaluates four models, each representative of distinct generative strategies: one unconditional GAN, one conditional GAN, and two conditional diffusion models. These were selected to assess whether model type and conditioning strategy affect the ability to synthesize T1c – a modality none of them have been trained to generate before. The following sections describe each model and the rationale behind its inclusion. Detailed model descriptions are available in Appendix 6.10.

3.2.1 *GAN Models*

The first is Pix2pixRAD, a conditional pix2pix GAN which synthesizes precontrast T1-weighted (T1w) brain MRI scans from T1c scans (Conte et al., 2021). This model produced high-fidelity scans that could successfully train a U-Net for a tumor segmentation task, even when trained on a small dataset. This model was chosen as its original training task involves the related but inverse mapping the present study is concerned with. Also, as a supervised conditional model, it was preferred over unsupervised frameworks like cycleGAN considering the present study used a complete dataset — allowing stronger training signal and potentially improved synthesis (Kong et al., 2021; Paavilainen et al., 2021).

StyleGAN2-ADA is an unconditional GAN which has been used to generate high-quality T1w scans of meningioma, glioma, and pituitary tumors using a small dataset and pretrained weights from non-medical domains (Karras et al., 2020; Tariq et al., 2023). It was also used to generate high-fidelity T1w healthy scans (Lai et al., 2025), indicating a high degree of adaptability in brain MRI. Notably, the successful use of pretrained weights in glioma scan generation demonstrates domain flexibility that may support generalization to T1c. Additionally, the study's use of unprocessed image data leaves generalization to standardized preprocessed datasets an open question. Furthermore, its performance has not yet been measured against diffusion models.

3.2.2 *Diffusion Models*

The first model, SynDiff (Özbey et al., 2023), performs unsupervised image-to-image translation across multiple brain MRI modalities, as well as pelvic CT images. Its unsupervised architecture, which allows for flexible modality translation without strict image pairing, together with its strong performance compared to other GAN and diffusion models, makes it a good candidate for cross-modality T1c synthesis — though its performance on downstream segmentation is unknown.

The second model is the custom diffusion model proposed by Mirowski and Fabijańska (2026) – henceforth referred to as custom diffusion — which generates FLAIR images conditioned on modified brain lesion segmentation masks. This represented a distinct strategy, since typical translation approaches involve conditioning on other brain scans. This model was selected due to its high performance in a segmentation task, and because its alternative conditioning strategy adds an additional dimension to the evaluation of how conditioning mechanisms affect T1c synthesis — namely, can T1c be synthesized from tumor spatial priors alone. However, the GAN benchmarks used in the study were not SOTA, making its performance compared to modern GAN and diffusion architectures unknown.

3.3 *Evaluating Synthetic Image Fidelity*

Synthetic brain MRI quality is commonly evaluated using image fidelity metrics spanning three domains. Distributional metrics, like Frechet Inception Distance (FID) and Kernel Inception Distance (KID) measure how well the synthetic image distribution matches that of genuine scans (Bińkowski et al., 2021; Heusel et al., 2018); structural and pixelwise metrics, like Structural Similarity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR), assess one-to-one and anatomical accuracy between synthetic and real scans (Wang et al., 2004); and perceptual metrics, like Learned Perceptual Image Patch Similarity (LPIPS) and Gradient Magnitude Similarity Deviation (GMSD) estimate visual similarity between scans as experienced by human observers (Xue et al., 2014; Zhang et al., 2018). Since no single metric can encompass all dimensions, a combined approach is necessary for holistic fidelity assessment (Deo et al., 2025; Dohmen et al., 2025). Therefore, this study will employ all aforementioned metrics in its fidelity analysis.

3.3.1 *Synthetic Images in Downstream Segmentation Tasks*

With the continued improvement of synthetic MRI generation, studies have begun to assess their utility for downstream tasks. For instance, Usman Akbar et al. (2024) demonstrated synthetic images from four GANs and one diffusion model could successfully be used to augment the training data of a segmentation model, with performance reaching 80-90% of real images alone. Additionally, the BraTS 2023 Adult Glioma segmentation challenge was won using GAN-based synthetic augmentation (Ferreira et al., 2024), underscoring the potential downstream clinical utility of synthetic data.

Researchers have also begun addressing the problem of scan missingness by testing substitution of missing MRI modalities with synthetic scans at inference time. The BraTS 2023 BraSyn challenge Li et al. (2024) aimed to benchmark this directly, considering segmentation pipelines require complete input data in order to be trained successfully but one or more scan types are commonly unavailable. Of the synthesis methods evaluated, GAN-based pipelines have consistently performed well (Baltruschat et al., 2024; Jain et al., 2026). However, T1c synthesis in particular remains challenging (Raut et al., 2024).

Raut et al. (2024) found T1c substitution to perform poorly compared to other modalities. The authors reasoned this was due to the inherently challenging nature of estimating gadolinium enhancement patterns from modalities without contrast information – explaining the particularly poor segmentation of enhancement-dependent tumor subregions. Similar challenges in reliably synthesizing T1c segmentation support the idea that the difficulty lies in the highly heterogeneous nature of the contrast patterns (Dong et al., 2026; Eidex et al., 2026; Huang et al., 2022).

More recently, diffusion models have also been applied to help address the missing modality problem, leveraging various frameworks for cross-modality synthesis (Friedrich et al., 2025; Meng et al., 2024; Özbey et al., 2023). However, their utility in segmentation is largely unknown due to comparatively fewer studies relative to GANs, especially for T1c synthesis.

3.4 *Literature Gaps and Contribution*

Although significant progress has been made in brain MRI synthesis, it is far more common for studies to focus on the generation of non-contrast modalities, while T1c synthesis is comparatively underrepresented despite its clinical importance (M. S. K. Luu et al., 2025; Mirowski & Fabijańska, 2026; Özbey et al., 2023; Tariq et al., 2023). Synthesis of one type of scan makes generalizability to other scan types unknown, especially considering the unique properties of T1c (Eidex et al., 2025). Additionally, studies may benchmark against older generative models, leaving the performance of their models relative to more contemporary architectures unclear (Ibrahim et al., 2025; Mirowski & Fabijańska, 2026; Tariq et al., 2023).

A key limitation among brain MRI synthesis studies is frequent reliance on image fidelity alone to assess image quality, leaving their potential clinical use in downstream tasks untested (Wu et al., 2024). Furthermore, studies commonly evaluate fidelity along a single dimension — e.g., reporting metrics related to pixelwise structure without also reporting perceptual or distributional metrics (Dohmen et al., 2025). In particular, Dohmen et al. (2025) observed that SSIM and PSNR are widely used despite having known weaknesses, with SSIM having difficulty capturing image blurriness and PSNR being a poor indicator of perceptual distortions. Therefore, relying on metrics of one aspect of fidelity alone paints an incomplete and potentially misleading picture of synthetic image fidelity (Deo et al., 2025).

Very few studies test generated images in downstream segmentation (Raut et al., 2024; Usman Akbar et al., 2024; Wu et al., 2024), and among those that do, T1c substitution tends to yield the poorest segmentation performance compared to all other modalities, especially in contrast-dependent subregions like necrotic core and enhancing tumor (Raut et al., 2024). Furthermore, this has been conducted almost exclusively using GANs, with diffusion-based approaches only recently emerging (Dayarathna et al., 2024; Eidex et al., 2026). This leaves the broader application of diffusion models to T1c synthesis less well understood. Finally, generalizability to external datasets is not frequently tested, making it difficult to assess overfitting (Ibrahim et al., 2025; Suleman et al., 2025).

This study seeks to build on BraSyn-related work, using Raut et al. (2024) as a key baseline, who found T1c substitution to yield the poorest segmentation performance, particularly for enhancement-dependent tumor regions. In adapting four SOTA generative models to T1c synthesis – one unconditional and three conditional – this study evaluates whether different conditioning strategies and architectures can improve T1c synthesis, and its downstream segmentation performance. Specifically, it makes the following contributions: 1) GAN and diffusion architectures are adapted to T1c synthesis for the first time, evaluating model flexibility while seeking to address T1c missingness; 2) The inclusion of an unconditional model tests if T1c is a learnable distribution; 3) Image fidelity is evaluated along distributional, pixelwise, and perceptual dimensions, providing a more comprehensive assessment of the quality of synthetic images; 4) Synthetic T1c images from each conditioned model are evaluated in a segmentation task, where they supplement missing T1c. These are compared to the segmentation of real-world baselines: a complete test set (no missing T1c), half-missing T1c, and completely missing T1c; and 5) Fidelity and segmentation performance for conditional models are tested for generalizability on an external dataset.

4 METHODOLOGY

4.1 *Datasets*

This study primarily used the BraTS-2023 Adult Glioma dataset, an open-access dataset consisting of full MRI brain scans of pre-treatment adult glioma patients at various stages of illness (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2015). MRIs were collected using various scanners and protocols at a total of 19 different institutions located across Germany, Hungary, India, Switzerland, and the United States (Bonato et al., 2025). Particular scanner models and associated hardware are not enumerated, and tumor type/grade (e.g., grade IV, HGG) were not made available.

The current study only used the provided training set, as the validation set did not contain segmentation masks. The training set consisted of 1,251 complete scans, each scan folder containing T1-weighted pre-contrast (T1w), T1 post-contrast (T1c), T2-weighted (T2w), T2-weighted FLAIR (T2f) scans, and their corresponding segmentation masks. All brain scan images were preprocessed (e.g., skull-stripped), and segmentation masks were expertly annotated. Segmentation mask label values correspond to pixel values representing the following sub-regions: enhancing tumor, where the blood-brain barrier has been broken down due to tumor growth (ET – pixel value 3); Peritumoral edema, or infiltrated tissue (ED – pixel value 2); and necrotic tumor core, containing dead or non-functioning tissue at the center of the tumor (NCR – pixel value 1). All other pixels were 0.

To evaluate generalizability this study also employed the University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) dataset (Calabrese et al., 2023). For brevity, this study will refer to the dataset as UCSF. This dataset includes complete preoperative brain scans of 501 glioma patients (6 of whom were follow-up) and includes several other scans in addition to those made available in the BraTS 2023 dataset. Data was collected between 2015–2021 at one medical center, and all MRI scans were performed with a single scanner type. Tumor types consisted of 55 patients (11%) of grade II, 42 patients (9%) of grade III, and 403 patients (80%) of grade IV tumor diagnosis.

4.1.1 *Dataset Preprocessing*

BraTS scans were downloaded as NIfTI files of dimension (240 x 240 x 155). 155 2D axial slices were generated at the scan level of dimension (240 x 240). These were stored as .png files, and subsequently resized to (256 x 256) with pixel intensity range [0, 255]. Slices were pruned at the scan level such that slices which contained $\leq$ 5% brain matter were removed, a similar method to the one employed by Mirowski and Fabijańska (2026) in order to ensure anatomical information. Subsequently, 15 of the remaining slices were selected at evenly spaced intervals in order to include clinically relevant brain regions. This was primarily done due to resource constraints, as training all model configurations on all slices was computationally prohibitive. However, 100 axial slices containing brain tissue per patient were preprocessed to train the best performing SynDiff configuration in order to replicate the experimental conditions of Özbey et al. (2023).

Data were split at the scan level into train and test sets (1000 : 251) according to the unstratified ratio used in Raut et al. (2024). However a small, hold-out validation set was created post-hoc by setting aside 10% of the training set in order to accommodate hardcoded training specifications in select models (train = 900: validation = 100, test = 251). Only StyleGAN2-ADA was trained on the full training set (80%). Data were further preprocessed according to the specific needs of each model (Figure 1).

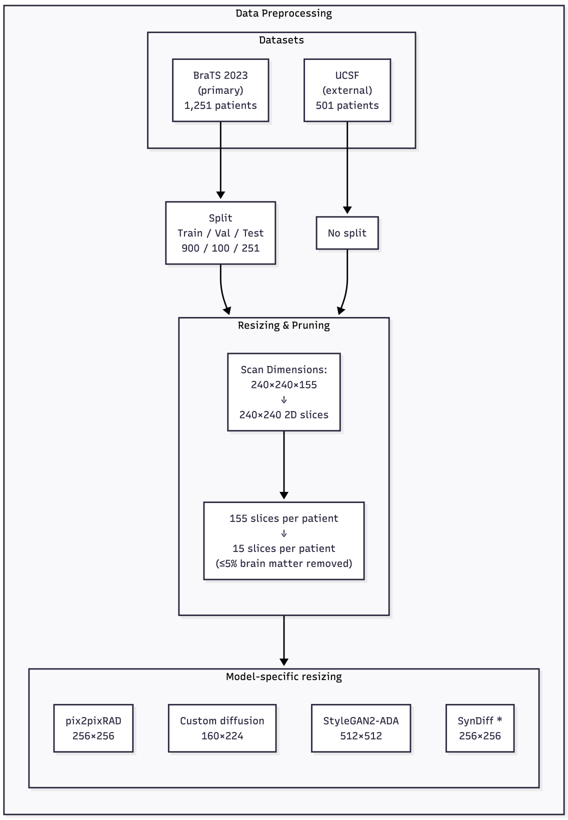

Figure 1: Preprocessing pipeline for BraTS and UCSF datasets. * The best model configuration for SynDiff was trained on 100 axial slices per patient, which was pruned in the same manner.

Data for pix2pixRAD were kept at (256 x 256), but were transformed to 3-channel RBG to match the expected input, and pixel intensities were normalized to [-1,1]. Additionally, data were augmented using random cropping and horizontal flipping. In order to make use of the pretrained weights for StyleGAN2-ADA training, data were resized to (512 x 512) and normalized to [-1,1]. SynDiff inputs were kept at the original resolution, but were normalized to [-1,1] and stored in .mat files. Finally, data for the custom diffusion model were center cropped to (160 x 224) with the target scans normalized initially to [0,1]. Scaled segmentation masks had pixel values scaled upwards to [0,255]. Modified segmentation masks were created by including healthy brain tissue at a pixel value of 4, with the resulting mask then scaled upwards to [0,255]. Figure 2 visualizes this process for clarity.

Scans from the UCSF dataset were processed into a single, unsplit dataset consisting of 495 unique patient scans and 6 follow-up scans. UCSF data preprocessing was identical to that of the BraTS dataset.

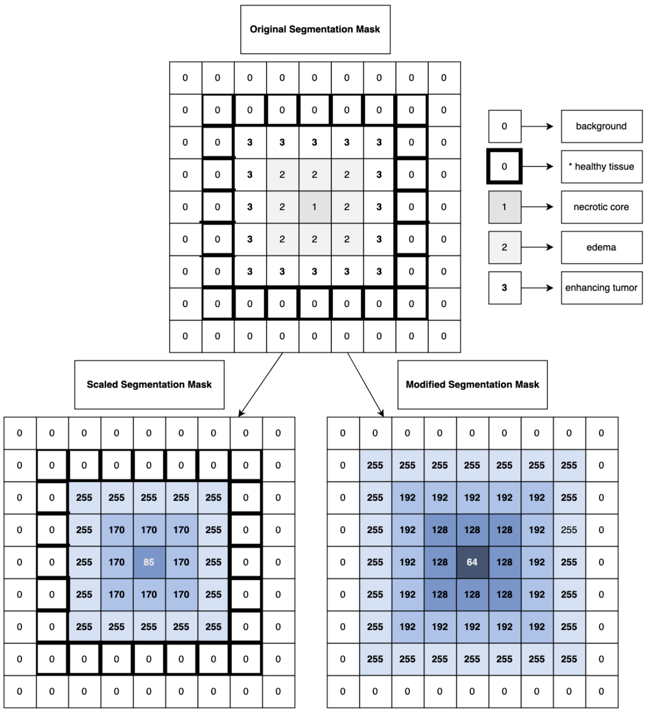

Figure 2: This figure depicts segmentation mask transformations. Tumor regions are denoted by pixel values 1-3, while 0 denotes background (* healthy brain tissue is shown as 0 with bold borders for additional clarity). Modified segmentation masks were made by setting healthy brain tissue to 4, then scaling to range [0,255]. Scaled segmentation masks applied the same scaling, but did not include healthy brain tissue.

4.2 *Environments*

All generative models were trained and evaluated using NVIDIA A40 GPUs (46GB VRAM, Driver 535.261.03, CUDA 12.2) provided via Tilburg University's GPU4EDU program. Each model was implemented in a dedicated conda environment, with dependencies shown in Appendix 6.10.

4.3 *Metrics*

4.3.1 *Image Fidelity*

FID and KID were used to evaluate distributional fidelity in all models – the former assessing the distance between the overall distributions and the latter reporting the mean of the distribution distance between individual images (Bińkowski et al., 2021; Heusel et al., 2018).

For conditional models, pixelwise similarity was assessed using PSNR, and structural learning using SSIM (Wang et al., 2004). Perceptual fidelity was measured with GMSD, which captures local edge-level differences (Xue et al., 2014); and LPIPS, which approximates human visual perception (Zhang et al., 2018).

GMSD, LPIPS, and SSIM values range from [0,1], with higher SSIM indicating better quality, and the reverse for the others. PSNR is measured in decibels (dB), with higher values representing closer pixelwise correspondence. KID and FID values do not have an upper bound, but lower FID values and KID values approaching 0 indicating high distributional similarity (Dohmen et al., 2025).

4.3.2 *Segmentation*

Tumor segmentation performance was gauged with the Dice Similarity Coefficient (Dice score), and the Intersection over Union (IoU) score, both measuring the degree to which two sets of data overlap. The range of both scores are (0,1), with a score of 1 indicating perfect overlap.

$$\mathrm{Dice}(A, B) = \frac{2\,|A \cap B|}{|A| + |B|} \qquad \mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Scores are reported for the mean of combined tumor regions, as well as the mean of each individual tumor region (ET, WT, TC). These are also reported for tumor sizes. Tumor size bins were constructed according to the total tumor pixels per patient statistic from the cross-validated (CV) nnU-Net. Small, medium, and large tumors were binned by percentiles, with small defined as below Q33, medium Q33 $\leq$ Q66, and large above Q66 (Table 1).

4.3.3 *Post-Hoc Error Analysis*

To verify image quality consistency between datasets, Signal-to-Noise Ratio (SNR) and Contrast-to-Noise Ratio (CNR) were calculated for each scan type per dataset. CNR is most directly associated with tissue differentiation, and SNR was included as a general measure of image quality (Magnotta & Friedman, 2006).

Table 1: Tumor size thresholds (in pixels) used to stratify patients into small, medium, and large categories for each subregion. Thresholds are defined as the 33rd and 66th percentiles of the training set pixel counts; the large category contains all cases at or above the 66th percentile.

	Pixels		
Size	*NCR*	*Edema*	*ET*
Small	254.95	2576.35	646.34
Medium	1285.48	6770.10	2276.84
Large	4955.91	13447.38	4895.94

Wasserstein's W1 was used to compare non-normal distributions between datasets, which measures dissimilarity between two distributions by evaluating the "cost" of moving data from distribution A to match distribution B (Villani, 2009). Significant differences between datasets were gauged using the two-tailed Mann-Whitney U test.

4.4 *Models*

Major architectural changes (e.g., adjustments to hidden layers) were considered out of the scope of the current experiment. Detailed training and model descriptions are available in Appendix 6.10.

4.4.1 *Pix2pixRAD*

Pix2pixRAD was selected to represent a conditional GAN architecture (Conte et al., 2021). Its original task (T1c $\rightarrow$ T1w) is the inverse of the goal of this study, and its supervised architecture complements the BraTS dataset, which contains complete scan pairs. Initial training following the baseline hyperparameter settings of Conte et al. (2021), which are described in Appendix 6.10. Binary cross-entropy loss (BCE) was replaced by least-squares loss (LSGAN) to prevent gradient saturation with discriminator loss (Mao et al., 2016). Perceptual (VGG-16) and gradient losses were added to encourage image sharpness and anatomical region differentiation (Johnson et al., 2016; Ma et al., 2020). Pretrained weights from the cityscape dataset were also employed to encourage initial structural learning (Cordts et al., 2016).

A core challenge of T1c synthesis was steering the model to generate correct contrast patterns and tumor anatomy. Tumor information was provided with a modified segmentation mask (scaled to [0,255]) concatenated to the second input channel. The final input configuration consisted of T1w, modified segmentation mask, and FLAIR (Figure 3. Full training description is reported in Appendix , with key hyperparameter adjustments.

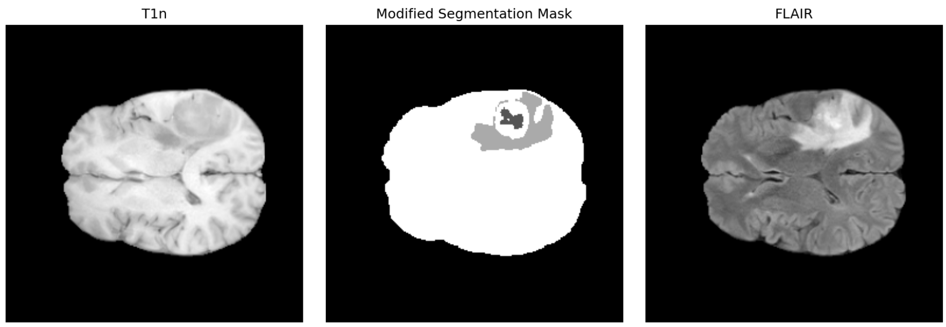

Figure 3: Sample of a 3-Channel input for pix2pixRAD, with T1w in the first channel, the modified segmentation mask (scaled to [0,255]) in the second, and FLAIR in the third.

4.4.2 *StyleGAN2-ADA*

StyleGAN2-ADA (Karras et al., 2020) was included as the sole unconditional model to test whether T1c is a learnable distribution, independent of conditioned synthesis strategies. The pretrained weights from the AFHQ wild 512 x 512 dataset were implemented during training (Choi et al., 2020), as Tariq et al. (2023) successfully used them to achieve competitive FID and KID scores. The generator at 1,880 kimg (i.e., thousands of images seen during training) was selected based on lowest FID score (see Appendix 6.10).

4.4.3 *SynDiff*

SynDiff was chosen as a conditional diffusion model due to its cross-modality translation without strict image pairing, potentially making it well suited to T1c synthesis from non-contrast modalities (Özbey et al., 2023). The best performing configuration was a modified T1w scan with embedded segmentation mask priors (Appendix A3: SynDiff). This was retrained on 100 axial slices per patient, as performed by Özbey et al. (2023) to evaluate model performance under less constrained data conditions.

4.4.4 *Custom Diffusion*

Custom diffusion (Mirowski & Fabijańska, 2026) was selected for its distinct conditioning strategy – generating scans from conditioned segmentation information rather than from other modalities. This adds an additional dimension to evaluation of differing conditioning strategies, and tests whether the already challenging T1c can be synthesized from limited spatial priors. The original implementation demonstrated modified segmentation masks were sufficient to synthesize FLAIR scans. This study attempted to apply this approach to T1c synthesis (see Appendix 6.12).

4.4.5 *nnU-Net*

The nnU-Net 2D model was chosen as the segmentation model against which synthetic T1c substitution was evaluated for its strong performance in brain tumor segmentation, and self-configuring architecture (Isensee et al., 2021; H. M. Luu & Park, 2021). The nnU-Net was trained on the full, unpruned training data. Best model weights were obtained through 5-fold cross validation.

4.5 *Workflow*

4.5.1 *Image Fidelity*

Following model training, all models generated 3,765 T1c scans (251 patients) for evaluation against the BraTS test set. All conditional models generated 7,515 T1c scans (501 patients) conditioned on the UCSF dataset to assess fidelity and segmentation performance generalizability against the external dataset. StyleGAN2-ADA was trained only on BraTS train data, and evaluated on distributional fidelity metrics for 3,765 generated images (Figure 4).

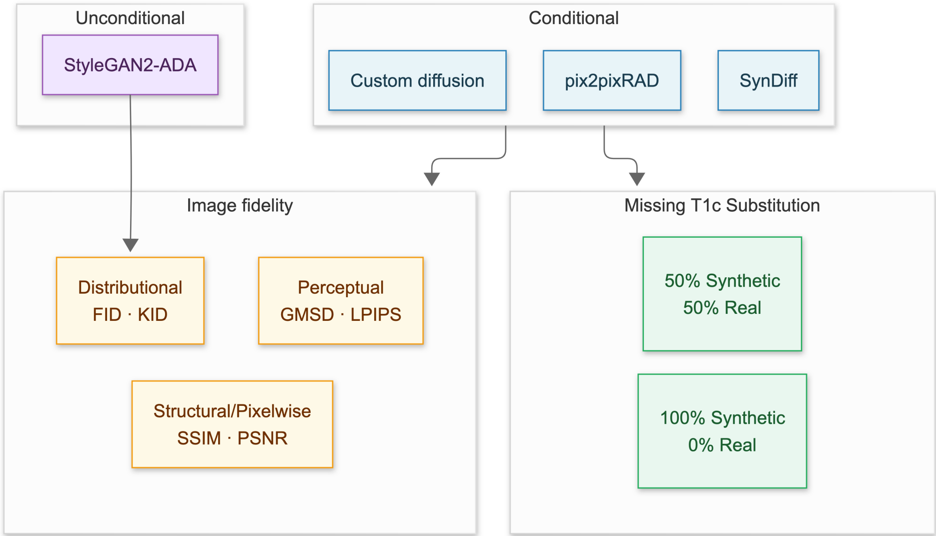

Figure 4: Synthetic image evaluation pipeline. Unconditional images are evaluated for distributional fidelity against BraTS test data only. Conditional images are evaluated on all three dimensions of image fidelity, and are evaluated as T1c substitutes in downstream segmentation. Conditional models generated synthetic T1c images conditioned on the UCSF dataset to evaluate generalizability. These models underwent the same fidelity and segmentation performance evaluation.

4.5.2 *Segmentation*

Segmentation was conducted over three composite tumor regions: Enhancing Tumor (ET), representing active tumor tissue; Tumor Core (TC), corresponding to ET and the necrotic core (NCR); and Whole Tumor (WT), consisting of all tumor subregions, including edema. Metrics were obtained per patient and per region, as well as averaged over all patients and regions.

Synthetic condition datasets consisted of the BraTS test set containing real T1w, T2w, FLAIR scans, as well as T1c scans that were either completely synthetic, or half-real, half-synthetic (see Figure 5). The half-synthetic split was done at the patient level to ensure the same split across all models. Performance of the synthetic conditions were compared to the cross-validated (CV) nn-UNet model which was trained on the complete, real BraTS train dataset, as well as to the real-world T1c missingness conditions. These were composed of the BraTS test dataset which contained complete real T1w, T2w, and FLAIR scans, as well as complete real T1c scans, half-missing T1c scans, or fully missing T1c. Scans in the missing conditions were replaced by images with uniform pixel value of 0.

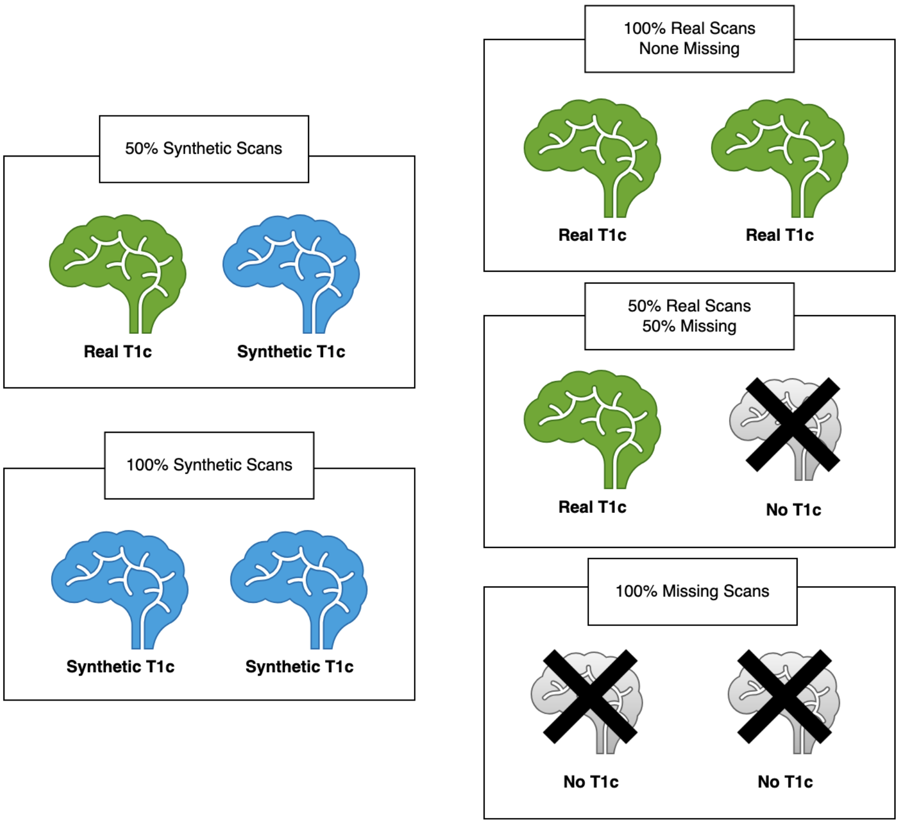

Figure 5: This image depicts the BraTS test set conditions for segmentation evaluation. The left conditions denote the synthetic T1c replacement conditions, where conditioned images replace either 50% or 100% of T1c in the BraTS test set. These are compared in segmentation performance to the conditions presented on the right, which represent real-world dataset missingness scenarios: the BraTS test data with 100% real T1c, 50% missing T1c, and completely missing T1c. These same conditions are applied to the UCSF dataset to examine generalizability of synthetic T1c as a replacement for missing T1c.

Predicted segmentation masks were generated from each experimental condition, and evaluated against ground truth segmentation masks. Segmentation results were stratified by tumor size. The same procedure was followed for the UCSF dataset except for tumor size stratification, which was deemed out of scope due to time constraints.

5 RESULTS

5.1 *Image Fidelity*

5.1.1 *Individual Model Performance*

Model checkpoints were selected based on best fidelity performance (see Appendix 6.10). For pix2pixRAD, the epoch 85 generator of the final configuration was selected, with Figure 6 depicting final and select intermediate model predictions.

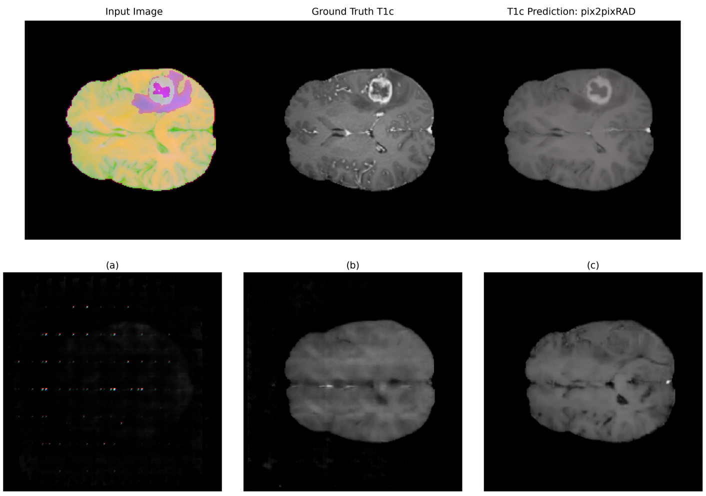

Figure 6: Examples of model outputs at various stages of training. The top right of the figure is the prediction of the best performing generator. The image on the top left is the input, consisting of a 3-channel image with T1w, a modified segmentation mask, and FLAIR. The top-center image is the ground truth T1c. Images labeled (a), (b), and (c) represent images generated using default hyperparameters and T1w input, adjusted hyperparameters with additional segmentation input, and final input configuration without LSGAN, respectively. Detailed training details are available in Appendix 6.10.

For StyleGAN2-ADA, the 1,880 kimg generator yielded the lowest FID (20.44) and was selected accordingly. Figure 7 shows a sample of final generator predictions.

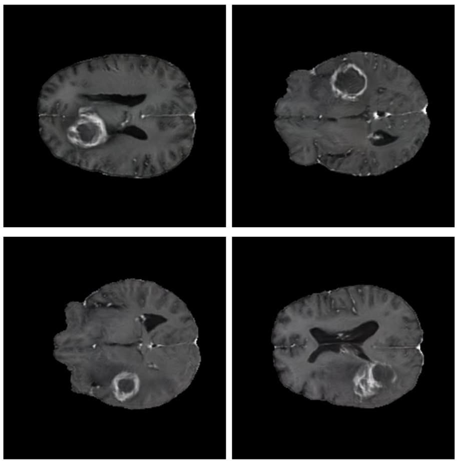

Figure 7: Sample images from the best-performing StyleGAN2-ADA generator.

For SynDiff, the T1w input model with segmentation intensity $\alpha = 0.3$ (Appendix A3: SynDiff) was selected for subsequent analysis. Samples of final and intermediate model predictions can be found in Figure 8.

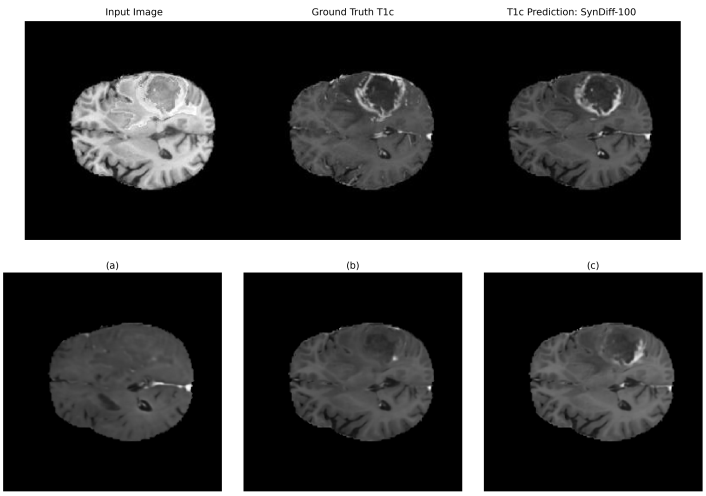

Figure 8: Examples of model outputs at various stages of training. The top right of the figure is the prediction of the best performing generator. The image on the top left is the input, consisting of T1w with a scaled segmentation overlay of intensity 0.3. The top-center image is the ground truth T1c. Images labeled (a), (b), and (c) represent images generated using conditioned FLAIR, T1w, and T1w with a 0.1 intensity segmentation overlay, respectively. Detailed training details are available in Appendix A3: SynDiff.

Finally, custom diffusion with baseline parameters was employed for the remainder of analyses. A sample of model predictions during training can be seen in Appendix 6.12.

5.1.2 *Cross-Model Comparison: RQ1, SQ1, and SQ2*

Table 2 and 3 refer to the cross-model comparisons on the BraTS test set and generalization dataset (UCSF). Figures 9 and 10 depict final model predictions compared to ground truth scans of BraTS and UCSF.

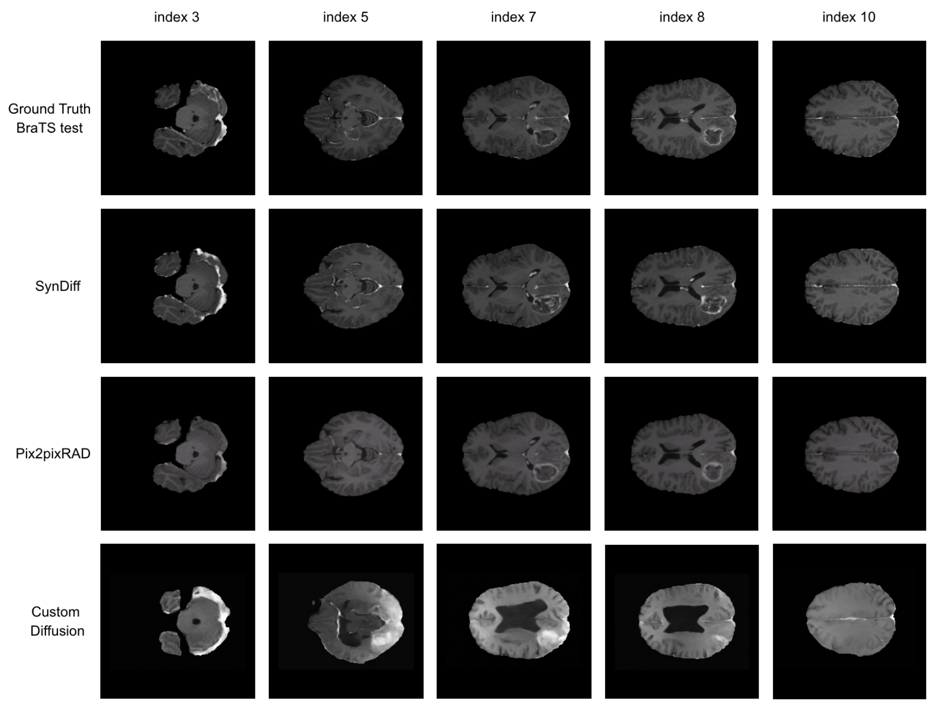

Figure 9: Sample axial slices of BraTS test set ground truth T1c, and their corresponding predictions from the three conditioned models. The columns represent the index of the slice within one patient scan (full index ranged 0-14). The rows represent the source of the images. Slices were chosen to represent glioma tumors as well as healthy scans.

SynDiff outperformed pix2pixRAD on image distribution metrics (FID: 18.52 vs. 37.17; KID: 0.0105 vs. 0.0260) on the BraTS test set, as well as the UCSF dataset (FID: 19.35 vs. 48.83; KID: 0.0122 vs. 0.0339). While SynDiff's image distribution scores degraded only slightly on the generalization set, pix2pixRAD's scores degraded notably more (FID: 37.17 $\rightarrow$ 48.83; KID 0.0260 $\rightarrow$ 0.0339).

StyleGAN2-ADA achieved the second lowest FID score (20.44), and outperformed SynDiff on KID, achieving the lowest score of 0.0060. Regard-

Table 2: Cross-model comparison of synthetic image fidelity using best model configurations (BraTS test-set).

Model	FID	KID	SSIM	PSNR	GMSD	LPIPS
Pix2pixRAD	37.17	0.0260	**0.9509**	**27.62**	**0.1171**	0.0634
StyleGAN2-ADA	20.44	**0.0060**	—	—	—	—
SynDiff	**18.52**	0.0105	0.9415	25.90	0.1219	**0.0592**
Custom Diffusion	97.51	0.0843	0.1061	11.70	0.3482	0.4324

Note: Cross model comparison of the best performing model configurations across image fidelity metrics for the BraTS dataset.

ing pixelwise fidelity, Pix2pixRAD outperformed SynDiff on both SSIM (0.9509 vs. 0.9415) and PSNR (27.62 vs. 25.90). However, this reversed with the UCSF dataset, with SynDiff achieving better scores than pix2pixRAD (SSIM: 0.9442 vs. 0.9409; PSNR: 27.61 vs. 26.26).

Table 3: Cross-model comparison of synthetic image fidelity using best model configurations, generalized to the UCSF dataset

Model	FID	KID	SSIM	PSNR	GMSD	LPIPS
SynDiff	**19.35**	**0.0122**	**0.9442**	**27.61**	0.1198	**0.0638**
Pix2pixRAD	48.83	0.0339	0.9409	26.26	**0.1164**	0.0791
Custom Diffusion	67.66	0.0573	0.1608	15.57	0.2592	0.2572

Note: Cross model comparison of the best performing model configurations across image fidelity metrics for the UCSF dataset.

Finally, SynDiff achieved lower LPIPS scores than pix2pixRAD on both the BraTS test set (0.0592 vs. 0.0634) and the UCSF dataset (0.0638 vs. 0.0791). Pix2pixRAD initially achieved a lower GMSD score than SynDiff on BraTS (0.1171 vs. 0.1219), but was outpaced by SynDiff (0.1198 vs. 0.1164). However, both models saw a slightly improved performance. Custom diffusion performed the worst across all metrics for both datasets, but did see an overall improvement in scores for each metric on the generalization set.

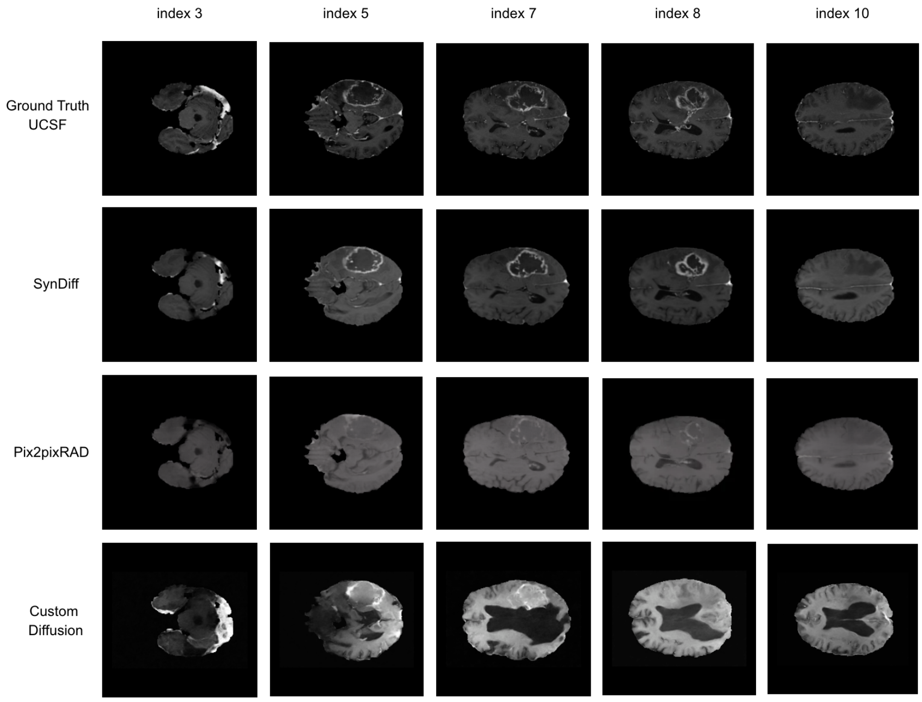

Figure 10: Sample axial slices of UCSF test set ground truth T1c, and their corresponding predictions from the three conditioned models. The columns represent the index of the slice within one patient scan (full index ranged 0-14). The rows represent the source of the images. Slices were chosen to represent glioma tumors as well as healthy scans.

5.2 *Segmentation*

5.2.1 *Real Data Performance: BraTS Test Set and UCSF*

Dice and IoU scores for direct BraTS test set and UCSF dataset comparison are reported in Table 4. Comparisons refer to Dice scores alone for brevity. Baseline nnU-Net performance on the complete training set after 5-fold cross validation (CV) yielded a mean Dice Score of 0.8599, and the following subregion Dice scores: 0.8261 (TC), 0.8685 (WT), and 0.8852 (ET). Performance was observed to degrade on the test set, with an overall mean Dice of 0.7357 and corresponding subregion scores of 0.6754 (TC), 0.7490 (WT), and 0.7826 (ET). The region with the greatest degradation in segmentation performance was TC, followed by WT, and ET.

No such degradation occurred in the UCSF segmentation performance in TC (0.840), WT (0.833), and ET (0.845) – all of which closely matched CV. However, differences in test and UCSF set performance diminished in the missing T1c conditions (50% missing: 0.479 vs. 0.524; 100% missing: 0.273

vs. 0.238). The WT subregion remained the most resistant to performance degradation, even under the 100% missing condition (BraTS: 0.608; UCSF: 0.645). The other subregions were severely affected by T1c missingness.

Table 4: Comparison of the UCSF dataset and BraTS test set segmentation performance per tumor subregion.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
cross-validated	**0.8599**	**0.8261**	**0.8685**	**0.8852**	**0.8057**	**0.7709**	**0.8109**	**0.8354**
BraTS test set: Real T1c								
0% missing	**0.7357**	**0.6754**	**0.7490**	**0.7826**	**0.6652**	**0.6110**	**0.6700**	**0.7145**
50% missing	0.4794	0.3471	0.6719	0.4192	0.4138	0.3037	0.5752	0.3625
100% missing	0.2729	0.0760	0.6075	0.1353	0.2106	0.0515	0.4914	0.0889
UCSF test set: Real T1c								
0% missing	**0.8396**	**0.8403**	**0.8333**	**0.8450**	**0.7828**	**0.7902**	**0.7687**	**0.7894**
50% missing	0.5236	0.4191	0.7415	0.4102	0.4749	0.3933	0.6562	0.3753
100% missing	0.2379	0.0114	0.6454	0.0567	0.1936	0.0075	0.5382	0.0351

Note: The nnU-Net model was trained on exclusively real, complete data. Performance of the test datasets are evaluated under complete T1c, partial missing, and complete missing T1c conditions. Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively.

Tumor segmentation performance in the BraTS test set varied with respect to tumor size, with large tumors having the highest mean Dice scores (CV = 0.9089, test = 0.8167), followed by medium sized tumors (CV = 0.8773, test = 0.7956), and small tumors (CV = 0.7580, test = 0.6646). The performance drop for the test set from medium to small was of a much larger magnitude than from large to medium (Δ = 0.131 vs. 0.021). WT was the most stable subregion. While mean Dice scores for TC in the test dropped by 0.5994 and ET by 0.6473 going from 0% to 100% missing , WT depreciated minimally (0.7490 $\rightarrow$ 0.6075). This was observed across all conditions and sizes (see Tables 6, 7, and 8). No tumor size analysis was conducted for UCSF due to time constraints.

5.2.2 *Synthetic Data Performance: RQ2*

In both pix2pixRAD and SynDiff, the 50% synthetic condition achieved much higher Dice scores (pix2pixRAD: 0.7011; SynDiff: 0.6770) than the

50% missing (0.4794) and 100% missing (0.2729) T1c conditions. 100% synthetic T1c datasets for both models also outperformed both missing T1c conditions (pix2pixRAD: 0.6789; SynDiff: 0.6294). While custom diffusion performed poorly on all regions, it nonetheless outperformed all missing conditions in the 50% synthetic condition. Furthermore, pix2pixRAD outperformed SynDiff in both the 50% (0.7011 vs. 0.6770) and 100% (0.6789 vs. 0.6294) synthetic conditions. Details are available in Table 5.

Table 5: Segmentation performance per model on the BraTS 2023 dataset across tumor subregions.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
5-Fold cross-validation	**0.8599**	**0.8261**	**0.8685**	**0.8852**	**0.8057**	**0.7709**	**0.8109**	**0.8354**
BraTS test set: Real T1c								
0% missing	**0.7357**	**0.6754**	**0.7490**	**0.7826**	**0.6652**	**0.6110**	**0.6700**	**0.7145**
50% missing	0.4794	0.3471	0.6719	0.4192	0.4138	0.3037	0.5752	0.3625
100% missing	0.2729	0.0760	0.6075	0.1353	0.2106	0.0515	0.4914	0.0889
BraTS test set: 50% Synthetic, 50% Real T1c								
Pix2pixRAD	**0.7011**	0.6605	**0.7351**	**0.7077**	**0.6256**	0.5910	**0.6524**	**0.6334**
SynDiff 100	0.6770	**0.6606**	0.7151	0.6554	0.6004	**0.5934**	0.6288	0.5791
Custom Diffusion	0.5424	0.4161	0.6648	0.5463	0.4652	0.3578	0.5755	0.4624
BraTS test set: 100% Synthetic T1c								
Pix2pixRAD	**0.6789**	**0.6609**	**0.7246**	**0.6513**	**0.5996**	**0.5889**	**0.6384**	**0.5715**
SynDiff	0.6294	0.6463	0.6931	0.5489	0.5465	0.5775	0.5984	0.4638
Custom Diffusion	0.3784	0.1929	0.5965	0.3457	0.2954	0.1413	0.4940	0.2508

Note: Classes 1, 2, 3 correspond to the mean segmentation performance on the tumor core (TC), whole tumor (WT), and enhancing tumor (ET) regions respectively. Mean refers to the averaged performance metric across all three tumor regions.

5.2.3 *Tumor Sizes and Subregions: SQ3*

All tumor size comparisons refer to Tables 6, 7, and 8. In both synthetic conditions, pix2pixRAD and SynDiff performed least well on ET relative to the 0% missing T1c condition (ET = 0.7826). Furthermore, performance dropped most significantly in ET going from 50% to 100% synthetic (pix2pixRAD: 0.7077 → 0.6513; SynDiff: 0.6554 → 0.5489) compared to much smaller drops in TC and WT.

Table 6: Segmentation performance for large tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
cross-validated	0.9089	0.8938	0.9148	0.9208	0.8649	0.8466	0.8703	0.8778
BraTS test set: Real T1c								
0% missing	**0.8167**	**0.7513**	**0.8359**	**0.8629**	**0.7506**	**0.6911**	**0.7627**	**0.7980**
50% missing	0.5244	0.3796	0.7611	0.4324	0.4570	0.3345	0.6625	0.3741
100% missing	0.3203	0.1078	0.7043	0.1487	0.2510	0.0751	0.5825	0.0955
BraTS test set: 50% Synthetic, 50% Real T1c								
Pix2pixRAD	0.7716	0.7248	**0.8262**	**0.7639**	0.6993	0.6598	**0.7486**	**0.6895**
SynDiff	0.7728	**0.7471**	0.8083	0.7631	0.6968	**0.6847**	0.7278	0.6780
Custom Diffusion	0.6112	0.4461	0.7663	0.6213	0.5288	0.3866	0.6769	0.5230
BraTS test set: 100% Synthetic T1c								
Pix2pixRAD	0.7523	0.7326	**0.8172**	**0.7070**	0.6759	0.6654	**0.7360**	**0.6263**
SynDiff	0.7498	**0.7594**	0.7908	0.6993	0.6667	**0.6950**	0.7032	0.6020
Custom Diffusion	0.4645	0.2303	0.6996	0.4636	0.3701	0.1729	0.5948	0.3427

Note: Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for large tumors are shown.

Performance decreased going from large to small tumors, and the same performance in ET relative to other regions was observed for each tumor size. Across all conditions and sizes, WT was the most stable region. While mean Dice scores for TC in the test dropped by 0.5994 and ET by 0.6473, WT depreciated minimally (0.7490 → 0.6075). This stability was such that even pix2pixRAD in the 100% synthetic condition had a similar Dice score to the unabated condition for WT (0.7246 vs. 0.7490).

Table 7: Segmentation performance for medium tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
cross-validated	0.8773	0.8463	0.8824	0.9033	0.8228	0.7882	0.8261	0.8541
BraTS test set: Real T1c								
0% missing	0.7956	**0.7526**	**0.8053**	**0.8289**	0.7232	**0.6866**	**0.7256**	**0.7573**
50% missing	0.5607	0.4231	0.7373	0.5219	0.4913	0.3766	0.6414	0.4560
100% missing	0.3006	0.0684	0.6424	0.1911	0.2289	0.0446	0.5165	0.1255
BraTS test set: 50% Synthetic, 50% Real T1c								
Pix2pixRAD	0.7673	0.7403	**0.7898**	**0.7718**	0.6891	0.6681	**0.7064**	**0.6927**
SynDiff	0.7649	**0.7432**	0.7820	0.7695	0.6835	**0.6695**	0.6959	0.6851
Custom Diffusion	0.6207	0.4994	0.7296	0.6330	0.5393	0.4348	0.6415	0.5416
BraTS test set: 100% Synthetic								
Pix2pixRAD	0.7376	0.7258	**0.7795**	**0.7076**	0.6551	0.6527	**0.6906**	**0.6219**
SynDiff	0.7219	**0.7428**	0.7491	0.6739	0.6280	**0.6639**	0.6515	0.5686
Custom Diffusion	0.4280	0.2291	0.6388	0.4161	0.3326	0.1671	0.5302	0.3004

Note: Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for medium tumors are shown.

Additionally, custom diffusion in the 100% synthetic condition achieved a higher Dice score for ET than the 100% missing baseline (0.3457 vs. 0.1353). Aforementioned observations regarding Dice scores between tumor regions also held for the stratified tumor sizes.

Table 8: Segmentation performance for small tumors across model variants.

	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
cross-validated	0.7580	0.6724	0.7941	0.8076	0.6890	0.6064	0.7153	0.7454
BraTS test set: Real T1c								
0% missing	0.6646	**0.6189**	**0.6517**	**0.7233**	0.5821	**0.5452**	**0.5560**	**0.6450**
50% missing	0.4774	0.3514	0.5758	0.5050	0.4016	0.3018	0.4680	0.4350
100% missing	0.2535	0.0615	0.5265	0.1726	0.1904	0.0399	0.4113	0.1201
BraTS test set: 50% Synthetic, 50% Real T1c								
Pix2pixRAD	0.6200	0.5975	**0.6288**	**0.6336**	0.5386	0.5236	**0.5313**	**0.5609**
SynDiff	0.5989	**0.6231**	0.5935	0.5801	0.5132	**0.5478**	0.4890	0.5027
Custom Diffusion	0.4875	0.4051	0.5385	0.5190	0.4084	0.3436	0.4379	0.4436
BraTS test set: 100% Synthetic T1c								
Pix2pixRAD	0.5742	0.5720	**0.6140**	**0.5366**	0.4911	0.4937	**0.5145**	**0.4650**
SynDiff	0.5308	**0.5799**	0.5644	0.4480	0.4373	**0.4998**	0.4530	0.3591
Custom Diffusion	0.2836	0.1262	0.4717	0.2528	0.2114	0.0884	0.3685	0.1772

Note: Classes 1, 2, 3 correspond to tumor core (TC), whole tumor (WT), and enhancing tumor (ET) respectively. Cross-validated BraTS training data are compared to segmentation of BraTS test data under real, 50% synthetic, and 100% synthetic T1c conditions. The total segmentation mean and subregion means for small tumors are shown.

5.3 *Segmentation Generalization: SQ4*

In the UCSF segmentation condition, Pix2pixRAD and SynDiff again outperformed both ablated conditions (pix2pixRAD: 0.6480; SynDiff: 0.6824). Compared to SynDiff, Pix2pixRAD had a notable depreciation of Dice scores across all tumor regions such that SynDiff had the best overall segmentation performance (Table 9). This was most pronounced in the ET region, where pix2pixRAD suffered a 22% drop in Dice score between the BraTS and UCSF dataset (0.7077 $\rightarrow$ 0.5494). As with BraTS, WT remained resistant to depreciation, with scores even marginally improving across both models.

Fully synthetic SynDiff outperformed the half synthetic condition (0.7023 vs. 0.6824), also outperforming both pix2pixRAD conditions(50%:

0.6480; 100%: 0.4482). There was an improvement in TC segmentation (0.6614 → 0.7326), while WT and ET remained largely stable between full- and half-synthetic SynDiff conditions. This was not observed in pix2pixRAD, where the expected depreciation in Dice scores occurred across all regions (0.6480 → 0.4482). Custom diffusion again performed the most poorly in both conditions, though still performing better than 100% missing T1c.

Table 9: Segmentation performance on the UCSF dataset per model across tumor subregions.

Model	Dice				IoU			
	Mean	1	2	3	Mean	1	2	3
BraTS training set								
cross-validation	**0.8599**	**0.8261**	**0.8685**	**0.8852**	**0.8057**	**0.7709**	**0.8109**	**0.8354**
UCSF dataset: Real T1c								
0% missing	**0.8396**	**0.8403**	**0.8333**	**0.8450**	**0.7828**	**0.7902**	**0.7687**	**0.7894**
50% missing	0.5236	0.4191	0.7415	0.4102	0.4749	0.3933	0.6562	0.3753
100% missing	0.2379	0.0114	0.6454	0.0567	0.1936	0.0075	0.5382	0.0351
UCSF dataset: 50% Synthetic, 50% real T1c								
Pix2pixRAD	0.6480	0.6229	0.7715	0.5494	0.5866	0.5737	0.6948	0.4914
SynDiff	**0.6824**	**0.6614**	**0.7765**	**0.6094**	**0.6240**	**0.6178**	**0.7004**	**0.5539**
Custom Diffusion	0.5779	0.5348	0.6927	0.5061	0.5133	0.4823	0.6157	0.4420
UCSF dataset: 100% Synthetic T1c								
Pix2pixRAD	0.4482	0.3974	0.7044	0.2428	0.3796	0.3464	0.6133	0.1792
SynDiff -	**0.7023**	**0.7326**	**0.7722**	**0.6022**	**0.6280**	**0.6759**	**0.6896**	**0.5185**
Custom Diffusion	0.3611	0.2426	0.5617	0.2790	0.2884	0.1874	0.4726	0.2053

Note: This table shows results from the segmentation generalization experiment, with UCSF used in place of BraTS. Synthetic data were used to supplement T1c from the UCSF dataset in the model conditions. Classes 1, 2, 3 correspond to the mean segmentation performance on the tumor core (TC), whole tumor (WT), and enhancing tumor (ET) regions respectively. Mean refers to the averaged performance metric across all three tumor regions.

5.4 *Error Analysis*

Post-hoc error analysis was conducted to investigate the segmentation discrepancy between the BraTS test set and the other datasets. The BraTS test set showed higher FP and FN rates, and more complete misses (Dice = 0) across tumor regions and sizes, especially in NCR and edema. Analysis uncovered elevated SNR and CNR statistics in T2w and FLAIR scans in catastrophic failures compared to other cases in all datasets. Furthermore, the BraTS test set had a substantially higher number of large tumors. Full results can be seen in Appendix 6.12.

6 DISCUSSION

This study investigated whether SOTA generative models could synthesize high-fidelity T1c scans that could substitute for missing T1c in downstream glioma segmentation. This was motivated by the frequent absence of T1c from clinical and research datasets in spite of its central importance for glioma assessment, particularly in HGG delineation. Evaluation of synthetic scans on fidelity and segmentation performance revealed T1c synthesis is achievable, but with substantial variation based on conditioning strategy, model type, tumor subregion and dataset. Overall, synthetic T1c substitution improved segmentation over missing T1c conditions, though not achieving performance of a complete dataset, a finding largely corroborated in prior work (Raut et al., 2024; Usman Akbar et al., 2024).

6.1 *RQ1: Can SOTA generative models generate T1c images that are comparable in quality to ground truth T1c*

Both pix2pixRAD and SynDiff achieved competitive fidelity across distributional, pixelwise and perceptual dimensions, suggesting high-fidelity T1c can be generated with appropriate conditioning. However, Initial model training revealed the difficulty of T1c synthesis, with frequent mode collapse and tumor learning failure observed in conditioned models when there was no tumor information present in non-contrast conditioned inputs (re: Appendix 6.10). In the case of pix2pixRAD, a multi-input conditioning approach with tumor spatial priors was ultimately successful, consistent with research that has found multi-input approaches to better handle contrast-dependent modalities (Dong et al., 2026; Kim & Park, 2024). Embedding tumor spatial priors into SynDiff's single input channel improved learning, analogous to other studies that have used segmentation masks as a conditioning input (Bhattacharya et al., 2025; Mirowski & Fabijańska, 2026). However, custom diffusion's failure to synthesize T1c with segmentation masks alone suggests spatial priors are insufficient for T1c conditioned synthesis, which is challenging enough using non-contrast scans (Dong et al., 2026; Eidex et al., 2025; Tabassum et al., 2024).

6.2 *SQ1: Can an unconditional model learn the T1c distribution?*

StyleGAN2-ADA performed strongly on distributional metrics compared to conditional models, indicating that T1c scans are learnable as a distribution independent of conditioning. This suggests that failures in conditional synthesis are due to insufficient guidance rather than a fundamental unlearnability of T1. Additionally, the model's distribution metrics compared favorably to scores reported in prior studies using it for non-contrast synthesis: the present study found FID = 20.44 with KID = 0.006, while Lai et al. (2025) reported FID ranging from 26.92 to 35.94, and KID ranging 0.026 to 0.038; and Tariq et al. (2023) reported FID = 58.1097 with KID = 0.009. This supports its potential downstream clinical utility, given the use of unconditional models in data augmentation (Motamed et al., 2021).

6.3 *SQ2: Can conditional models generalize to an external dataset?*

On the BraTS test set, SynDiff demonstrated better distributional and perceptual fidelity, consistent with evidence that diffusion models produce more diverse image distributions than GANs (Müller-Franzes et al., 2023); while pix2pixRAD had stronger pixel-wise learning, consistent with GANs exhibiting more rigid structural learning (Isola et al., 2018). These results explain the better generalizability of SynDiff to the UCSF dataset than pix2pixRAD, since the rigidity of pix2pixRAD's learning may have led to pronounced degradation in its distributional metrics. Since SynDiff's fidelity remained stable, this suggests superior distribution learning allowed for greater generalizability.

6.4 *RQ2: How well can synthetic conditional T1c images substitute for missing T1c in downstream segmentation?*

Synthetic T1c substitution with pix2pixRAD and SynDiff outperformed missing T1c conditions, demonstrating that SOTA diffusion and GAN models can be adapted to generate potentially clinically useful T1c scans. Performance remained below the real T1c condition, though this is consistent with the use of synthetic modalities in segmentation (Raut et al., 2024; Usman Akbar et al., 2024).

Pix2pixRAD outperformed SynDiff despite inferior distributional and perceptual fidelity, suggesting tumor region accuracy may matter more than holistic fidelity for downstream segmentation (Raut et al., 2024). This aligns with evidence that shows image quality metrics may be insensitive to local anatomical details critical for accurate segmentation (Deo et al., 2025).

Therefore, pix2pixRAD's rigid pixel-wise mapping may have produced tumor regions that initially better corresponded to nnU-Net expectations.

6.5 *SQ3: How does synthetic T1c segmentation performance vary by tumor size/region?*

Segmentation performance decreased as tumor regions became smaller, which is consistent with previous segmentation research (Erdur et al., 2023; Raut et al., 2024). Specifically, the underperformance of ET segmentation for all synthetic models revealed a key weakness in the T1c synthesis approaches: namely, models were unable to sufficiently learn tumor enhancement patterns central to the modality – even after conditioning with tumor spatial priors. This may be due to the high degree of heterogeneity of gadolinium enhancement (Osman & Tamam, 2023; Warntjes et al., 2018). Consequently, it represents potential disparate impact for use in segmentation supplementation in glioma patients – especially those who present with pronounced ET compared to other regions (e.g., HGG).

However, despite this decreased performance, both models improved on the Raut et al. (2024) baseline, which found the best performing T1c substitution condition achieved Dice = 0.12 (± 0.11) on the ET region (full replacement pix2pixRAD = 0.65; SynDiff = 0.55). Additionally, both pix2pixRAD and SynDiff outperformed missing T1c conditions with respect to the TC region. Because TC incorporates both NCR and ET, this suggests that the models were able to synthesize robust enough NCR regions to compensate for poor ET estimation. Therefore, despite difficulty with ET learning, synthetic T1c nonetheless may offer genuine clinical utility for non-enhancing tumor tissues. Incidentally, the stability of WT across all models and conditions was consistent with the known relationship between T2w/FLAIR and edema visualization (Stall et al., 2010), making WT segmentation performance less dependent on T1c presence.

6.6 *SQ4: Does synthetic T1c segmentation performance generalize to an external dataset?*

While both models again outperformed missing T1c conditions on UCSF, robustness was contingent on distributional flexibility rather than pixelwise correspondence. Pix2pixRAD showed notable deterioration across all tumor regions, consistent with the tendency of GANs of having difficulty generalizing due to overly rigid mappings (Isola et al., 2018). This was most pronounced in ET, demonstrating pixelwise learning could not fully capture heterogeneous enhancement patterns that vary widely between patients (Osman & Tamam, 2023; Warntjes et al., 2018).

Most surprisingly, 100% synthetic SynDiff outperformed 50% synthetic conditions on UCSF. This counterintuitive finding may reflect distributional dissimilarity, as previous research notes that performance can be influenced by distributional consistency as well as image fidelity (Jahagirdar & Joshi, 2025; Usman Akbar et al., 2024). Since UCSF data came from a single institutional scanner, BraTS-trained synthetic T1c was likely more distributionally similar to BraTS than UCSF. The segmentation model may have therefore encountered a distributional mismatch in the half synthetic condition that was not present in the full synthetic condition, leading to slightly reduced segmentation.

6.7 *Error Analysis*

While there was a notable tumor size class imbalance present in the BraTS test set, the train-test performance gap appeared to be driven primarily by a chance over-representation of atypical glioma phenotypes in the test set. Failed cases showed a higher concentration of elevated T2w and FLAIR signal metrics, which were overrepresented in the test set. This may indicate more extensive, diffuse edema obscuring other subregion boundaries (Claes et al., 2007; Schmitz-Abecassis et al., 2025; Sengupta et al., 2018). High FP rates in catastrophic failures confirmed other tumor regions were mislabeled as edema. Since diffuse glioma tumors may be challenging to delineate accurately, a higher concentration of such tumors is a plausible explanation for the performance discrepancy (Wan et al., 2024).

6.8 *Societal and Scientific Implications*

This study addressed key scientific gaps by adapting both conditional and unconditional GAN and diffusion architectures for T1c synthesis, and evaluating performance on a segmentation task with generalization. These gaps include: the under-representation of T1c in MRI synthesis (Dong et al., 2026; Eidex et al., 2025; M. S. K. Luu et al., 2025), incomplete image fidelity evaluation (Deo et al., 2025; Dohmen et al., 2025); assessment of image fidelity without also testing downstream clinical utility (Wu et al., 2024); the over-representation of GANs in downstream tasks compared to diffusion models (Dayarathna et al., 2025; Eidex et al., 2026); and limited assessment of generalizability (Ibrahim et al., 2025; Suleman et al., 2025).

The societal implications of this study involve the improvement of T1c access for glioma patients. T1c scarcity is widespread due to resource and time constraints (Kim & Park, 2024); and gadolinium contrast is inadvisable for particular patient groups, like pregnant women and those with kidney disease (Bird et al., 2019; Jayachandran Preetha et al., 2021; Schieda et al., 2018). Therefore, research towards a solution for T1c missingness would benefit patients with glioma who cannot obtain scans, but for whom T1c scans are crucial for assessment. While the results of this study cannot be directly deployed in clinical practice, they nonetheless provide a concrete step towards the generation of reliable, clinically useful T1c scans.

Considering ET delineation is critical for glioma assessment, especially in the case of highly malignant HGGs with poor prognosis (Hirschler et al., 2023; Unkelbach et al., 2014), the persistent failure of all models to accurately synthesize ET represents a meaningful limitation in clinical utility. The misapplication of such T1c synthesis models risks disparate negative impact on glioma patient care. Therefore, current synthetic T1c use may be better suited for non-enhancing tumor tissue segmentation, where more reliable performance was demonstrated.

6.9 *Limitations and Future Directions*

This study had several limitations. The initial train-test split was unstratified, which may have contributed to an over-representation of atypical glioma phenotypes in the test set. All models were trained on 2D axial slices rather than 3D volumes, limiting spatial context and direct clinical applicability. Image fidelity evaluation would have been strengthened with blind rating by radiologists; and segmentation assessment would have benefited by the inclusion of Hausdorff's distance, which captures worst-case boundary errors that Dice/IoU cannot measure (Aydin et al., 2021). Additionally, all data were heavily preprocessed, leaving performance on raw, unprocessed clinical data unknown. Finally, both datasets were representative of wealthier, western demographics, limiting generalizability to under-resourced settings where T1c scarcity is more acute.

Future studies should stratify by tumor size and grade (if labeled) to avoid chance imbalances in glioma type representation, and extend T1c conditioning strategies to 3D GAN and diffusion architectures. Additionally, research should be extended to diverse patient populations such as BraTS-Africa (Adewole et al., 2025), or to pediatric gliomas. Fine-tuning nnU-Net with mixed synthetic T1c data may improve evaluation performance without compromising performance on real scans (Pani & Chawla, 2024; Usman Akbar et al., 2024). Finally, mask-free conditioning and improved enhancement learning are identified as key steps for more complete synthetic T1c clinical utility in glioma patients.

6.10 *Conclusion*

T1c scans are critical for glioma assessment, yet frequently absent from clinical and research datasets, adversely affecting patient care and segmentation model use. This study demonstrated that SOTA GAN and diffusion models can be adapted to generate high-fidelity T1c scans that improved segmentation over missing T1c conditions. Though tumor information was necessary but not sufficient to guide synthesis, and enhancing tumor regions were consistently challenging to synthesize, these results serve as a meaningful proof-of-concept that synthesis is a generalizable solution to T1c scarcity. With improvements to conditioning strategy and enhancement learning, synthetic T1c generation could foreseeably expand access to crucial glioma assessment for patients unable to obtain contrast scans.

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APPENDIX A

Table 10: Software dependencies across models

Dependency	Pix2PixRAD	StyleGAN2-ADA	SynDiff	Custom Diffusion	nnU-Net
Python	3.10.0	3.8.20	3.8.20	3.10.19	3.10.19
PyTorch	2.10.0	1.7.1	1.7.1	2.0.1	2.5.1
TorchVision	0.25.0	–	0.8.2	0.15.2	0.20.1
TorchAudio	–	–	–	–	2.5.1
TensorFlow	2.20.0	–	–	–	–
Keras	3.12.0	–	–	–	–
CUDA wheel	–	11.0	11.0	–	12.1
CUDA Runtime	12.8.90	–	–	11.7.99	12.1.105
CUDA Toolkit	–	–	11.0.3	–	–
cuDNN	9.10.2.21	–	8.0.5	8.5.0.96	9.1.0.70
NumPy	2.2.6	1.24.4	1.24.3	1.26.4	2.2.6
SciPy	1.15.3	1.10.1	1.10.1	1.15.2	1.15.3
Pandas	–	–	2.0.3	–	2.3.3
Pillow	12.1.0	10.4.0	9.3.0	–	12.0.0
Matplotlib	–	–	3.7.5	–	3.10.8
Seaborn	–	–	0.13.2	–	0.13.2
scikit-image	–	–	0.21.0	–	0.25.2
scikit-learn	–	–	–	–	1.7.2
imageio	–	–	2.35.1	–	–
imageio-ffmpeg	–	0.4.3	–	–	–
h5py	–	–	3.11.0	–	–
MONAI	–	–	–	1.4.0	–
MONAI-Generative	–	–	–	0.2.3	–
NiBabel	–	–	–	5.3.3	5.4.0
SimpleITK	–	–	–	–	2.5.3
nnunetv2	–	–	–	–	2.6.4
pyspng	–	0.1.4	–	–	–
Ninja	–	1.13.0	1.13.1	–	–
click	–	8.1.8	–	–	–

Dependency	Pix2PixRAD	StyleGAN2-ADA	SynDiff	Custom Diffusion	nnU-Net
requests	–	2.32.4	–	–	–
tqdm	–	4.67.1	–	–	–

A1: Pix2pixRAD

ARCHITECTURE The current study employed the same baseline architecture as Conte et al. (2021). The Pix2pixRAD generator consists of 8 convolutional downsampling layers composing a single downstack, and 8 convolutional upsampling layers composing an upstack. The function of the downstack is to compress the input image into a compact feature representation to encourage the model to learn foundational structural features, whereas the upstack forces the model to reconstruct the image to its 256 x 256 resolution using the features learned from the downstack. The generator also makes use of batch normalization to stabilize training, skip layers to pass along feature information, and leakyReLU to avoid disappearing gradients by allowing small negative gradients. Figure 11 visualizes the basic generator architecture.

The discriminator architecture is composed of three successive downsampling layers, and likewise makes use of batch normalization, leakyReLU. In addition, zero padding is applied to ensure no edge information is lost between convolutions. A representation of the discriminator is shown in Figure 12. The model originally used binary cross entropy adversarial loss, training the discriminator to distinguish between real and synthetic images using the probability of synthetic vs. real. Finally, the model used an Adam generator and discriminator optimizer of 2e-4 and originally employed a λ L1 loss of 100, which functions to strongly enforce pixelwise similarity between the predicted and target image.

TRAINING Initial training was implemented for 300 epochs as specified by Conte et al. (2021) using the following hyperparameters: Binary cross entropy loss, λ-L1= 200, Adam optimizer with generator and discriminator loss = 2e-4, $\beta 1$ =0.5, $\beta 2$ =0.99, and minibatch stochastic gradient descent with batch size = 1. The initial model was trained in a translation task, from T1w $\rightarrow$ T1c, using input of T1w scan across 3 channels.

Model training was marked by frequent mode collapse, along with rapid discriminator loss decrease coupled with generator loss stagnation. The discriminator learning rate was incrementally decreased to 5e-5, mitigating mode collapse–however, outputs were consistently blurry and anatomically undifferentiated. To reduce over-reliance on penalizing pixel-

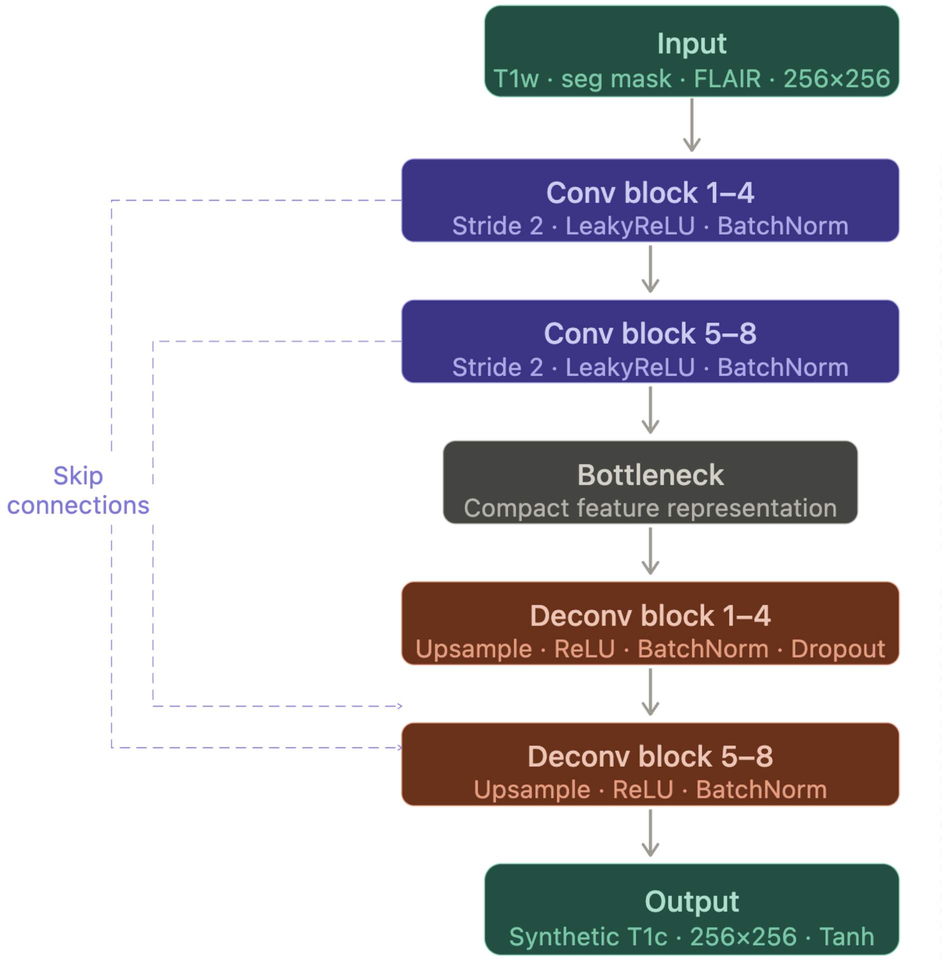

Figure 11: A representation of the pix2pix generator architecture.

wise differences and encourage the generator to produce a greater degree of contrast between brain regions, λ-L1 was decreased incrementally to 3. Though helpful, these changes did not help the model to sufficiently learn mappings to tumor regions.

Reasoning that T1w alone did not provide sufficient tumor-related information, the model was trained with a modified input, containing T1w in channels 1 and 3, and a segmentation mask scaled to [0, 255] in channel 2. Despite marginal improvements in image quality, outputs still lacked anatomical contrast. Therefore, perceptual loss (VGG-16) and gradient loss were introduced to encourage image sharpness and edge-level differentiation, since decreasing λ-L1 alone was insufficient (Johnson et al., 2016; Ma et al., 2020).

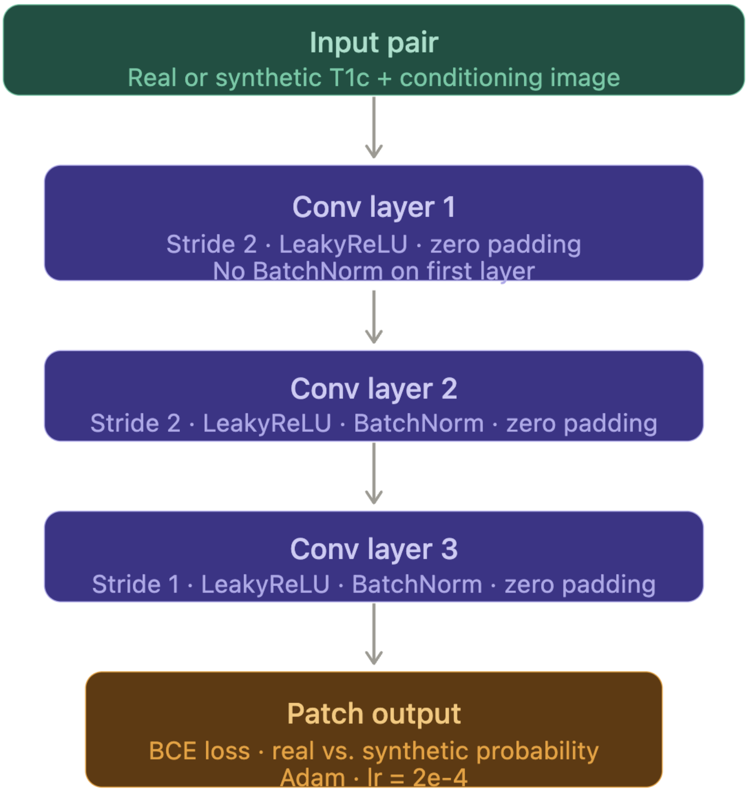

Figure 12: A representation of the pix2pix discriminator architecture.

To further stabilize training, pretrained weights were used from the Cityscape dataset (Cordts et al., 2016). In spite of the domain gap, this stabilized training dynamics significantly, though the model still suffered from blurred brain anatomy. Finally, the input was further modified such that the channels contained T1w, the scaled mask, and the corresponding FLAIR scan, respectively. This was done to encourage learning tumor-relevant information and anatomical detail that is not fully present in both T1w scans and segmentation masks. Additionally, binary cross-entropy loss was replaced with least-squares loss (LSGAN) (Mao et al., 2016).This was done to avoid saturated gradients common with binary cross-entropy loss, which resulted in discriminator loss consistently outpacing generator loss.

The final model consisted of pretrained cityscape weights, an input of T1w, scaled segmentation masks, and FLAIR. Furthermore, the following hyperparameters were implemented: least-squares loss, λ-L1 = 1, batch size = 2, generator loss = 2e-4, discriminator loss = 5e-5, β L1 = 0.5, λ-

perceptual = 10, λ gradient = 10, epochs = 85. A comprehensive overview of training is shown in Table 11.

Table 11: Pix2PixRAD hyperparameter configurations across training stages

Parameter	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7
Input Channels	T1w x 3	T1w x 3	T1w, seg, T1w	T1w, seg, T1w	T1w, seg, FLAIR	T1w, seg, FLAIR	T1n, seg, FLAIR
Epochs	300	–	–	–	–	–	85
Loss Function (BCE)	✓	✓	✓	✓	✓	✓	LSGAN
λ L1	100	25	3	3	3	1	1
λ Perceptual	–	–	–	–	10	10	10
λ Gradient	–	–	–	–	10	10	10
Generator LR	2e-4	2e-4	2e-4	1e-4	1e-4	1e-4	1e-4
Discriminator LR	2e-4	5e-5	5e-5	5e-5	5e-5	5e-5	5e-5
Batch Size	1	1	1	1	2	2	2
Pretrained Weights	–	–	–	Cityscape	Cityscape	Cityscape	Cityscape
Assessment	Mode collapse: black outputs; near zero discriminator loss	Mode collapse; grainy images with minimal anatomical differentiation	Reduced blurring; improved but poor anatomical learning.	Image quality improvement; still lacking sufficient anatomical contrast	Significant improvement of anatomy learning; images remain blurry	Stabilized training dynamics; improved image sharpness	Stable training; successful learning of tumor anatomy

Note: LR = Learning Rate; BCE = Binary Cross-Entropy; LSGAN = Least Squares GAN; Loss function was MSE unless otherwise specified; λ Perceptual = VGG-16; seg = modified segmentation mask. Epochs less than 50 were not specified, given early model failure. More configurations were tried besides those listed in the table, but adjustments were deemed minor enough to not warrant inclusion.

A2: StyleGAN2-ADA

ARCHITECTURE This study employed the same architecture as was used by Tariq et al. (2023). The StyleGAN2-ADA generator is composed of a mapping network with 8 fully connected layers, as well as a synthesis network with 26 modulated convolutional layers (Karras et al., 2020). The mapping layers are responsible for translating the raw latent space Z into a new latent space of W, which correspond to general image features. Importantly, the learning rate of the mapping network is set to 100x lower than the synthesis network (lrmul=0.01) to assure training stability, considering the significantly lower dimensionality of latent image representations.

The synthesis network consists of alternating upsampling and ToRGB layers, which are modulated by a vector from W (mapped from a sampled Z vector) rather than fixed weights, allowing the network to learn through individual images rather than a "one-size-fits-all" weight matrix. Weights from W are also normalized to prevent exploding gradients. The ToRGB connections function to generate a "partial image," allowing information to be more stably passed on the fully connected layers. In addition, gaussian noise is added to feature maps at each layer, thereby allocating the generation of global structures to W and the generation of fine-grained detail to injected noise. A representation of the mapping and synthesis networks are shown in Figure 13.

Figure 13: A representation of the StyleGAN2-ADA generator architecture.

The discriminator consists of 8 resolution blocks, consisting of 24 convolutional layers preceded by a single convolutional layer responsible for translating from 3-channel RGB image space to the discriminator feature space. It also consists of a minibatch standard deviation layer, which measures batch diversity, helping the discriminator detect mode collapse (Figure 14.

TRAINING While Tariq et al. (2023) implemented the model with TensorFlow, this study elected to use the PyTorch version due to dependency issues. Pretrained weights from the AFHQ wild 512 x 512 dataset were used (Choi et al., 2020), as Tariq et al. (2023) used these weights to train a

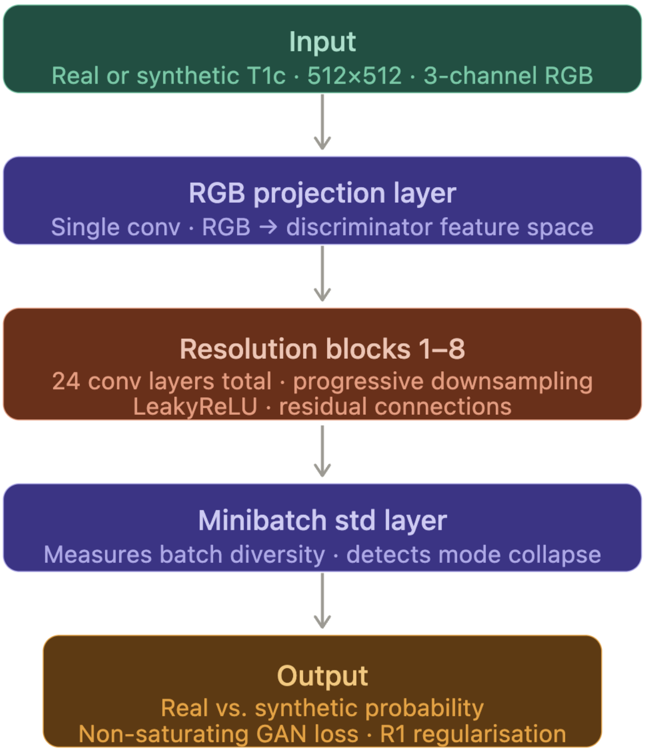

Figure 14: A representation of the StyleGAN2-ADA generator architecture.

model with the second best FID/KID values at a permissible resolution of 512 x 512.

The model was trained using the PyTorch implementation, and was run for a total of 2,260 kimg. The generator at 1,880 kimg was ultimately chosen, as it had the lowest calculated FID value. No hyperparameter adjustments were conducted.

6.11 *A3: SynDiff*

ARCHITECTURE The current study employed the same baseline architecture as Özbey et al. (2023). SynDiff employs a hybrid architecture, training both a non-diffusive and diffusive module jointly. The purpose of the non-diffusive module is to estimate an input image paired to each target image, in order to better guide the diffusion model. It consists of two pairs of GANs, the generators of which are built on a ResNet backbone, with 3 encoding blocks, 6 residual blocks, and 3 decoding blocks - an example of which can be seen in Figure 15. Both GANs are trained with adversarial loss, while also applying cycle-consistency loss to ensure that the model can learn to translate in both directions – i.e., A B A (Figure 16).

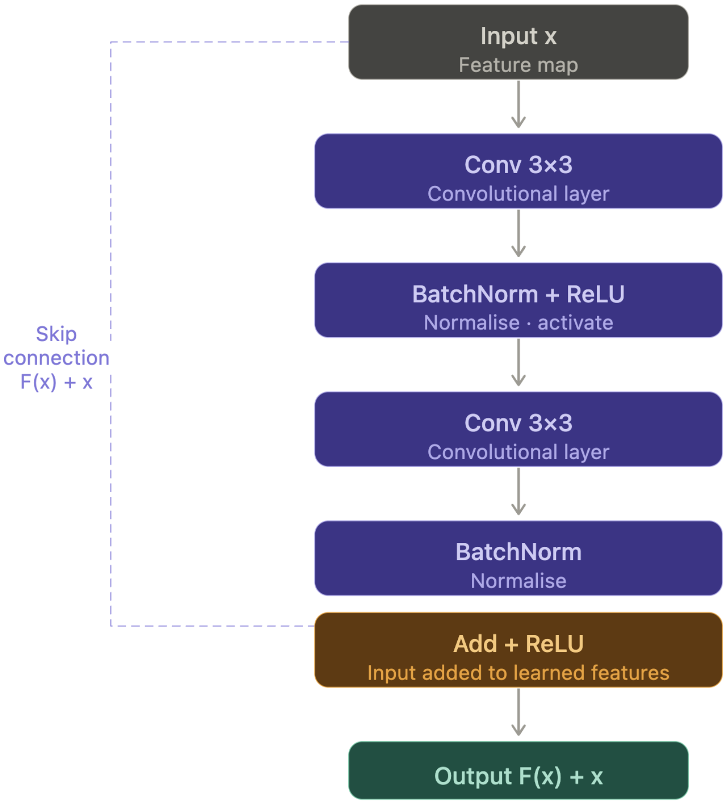

Figure 15: A representation of a ResNet architecture.

The diffusive module consists of two generators with an underlying UNet architecture consisting of 6 encoding and decoding blocks, as well as residual subblocks and temporal/latent embeddings to adjust the degree of applied noise based on the current position along the diffusion process. These are paired with discriminators which are comprised of 6 blocks of 2 convolutional layers with downsampling. The diffusion process comprises of 4 reverse diffusion steps, with 1000 steps and a step size of k = 250. An example of the diffusive module can be seen in Figure 17.

As the task of this study was to synthesize T1c scans in one direction, only the generator performing this unidirectional mapping was employed. By extension, only the relevant diffusion generator was used at inference.

TRAINING This study made use of the 50-epoch pretrained weights for the T2w $\rightarrow$ T1w translation task available on the SynDiff GitHub repo (Özbey et al., 2023). This was done both to save computing resources, and to take advantage of learned brain MRI mappings. Additionally, this

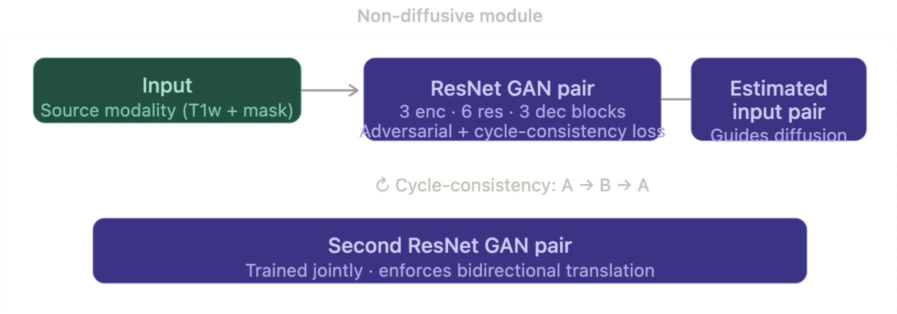

Figure 16: A representation of the SynDiff non-diffusive module architecture.

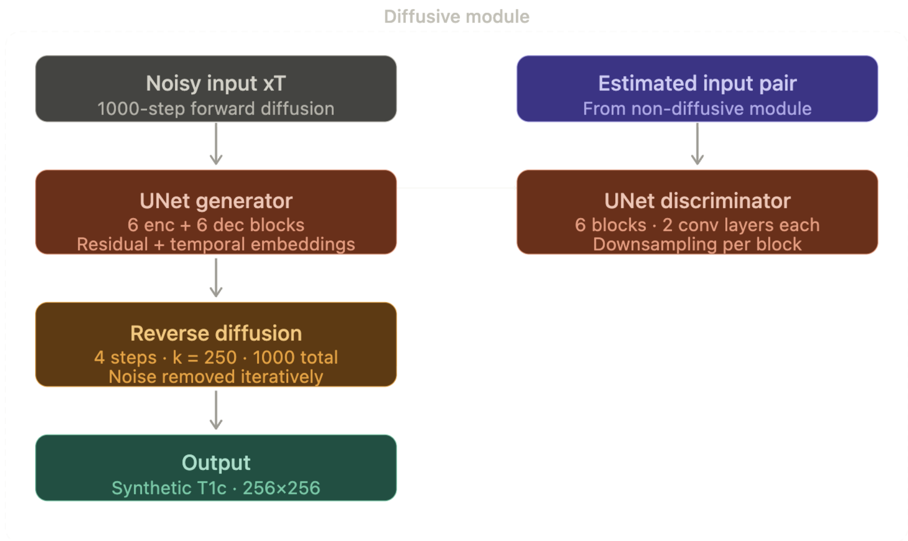

Figure 17: A representation of the SynDiff non-diffusive module architecture.

specific direction was chosen due to the relatedness of T1w and T1c scans, and it was reasoned that the pretrained model had learned to generate anatomical features consistent with the intensity patterns present in T1 scans. Batch size was initially increased from 1 to 2 to speed up and stabilize training.

In order to determine which scan type would provide the most useful anatomical information for a translation task to T1c, this study tested four different inputs: FLAIR, T1w, and two versions of a modified T1w scan. The modified scans consisted of T1w scans concatenated with a scaled segmentation mask ([0,3] $\rightarrow$ [0,255]) such that pixel intensities were transformed according to the following equation: modified T1w = T1w + α * scaled segmentation mask, with intensities of $\alpha = 0.1$, and $\alpha = 0.3$.

FLAIR was chosen due to its similarity to the original input of T2w scans, as well as the fact that it contained tumor-related information and rich anatomical detail. Though pixel intensity distributions differ widely between FLAIR and T1c, it was reasoned that they differ in similar enough ways from T2w and T1w to warrant testing. T1w was selected considering it is in a similar domain to T1c, so the model may have an easier time with similar input and target pixel intensity distributions. Finally, T1w scans modified with scaled segmentation masks were used as an exploratory test to see if the model would benefit from additional tumor related information that would otherwise not be present in T1w scans alone.

All models were initially trained for 5 epochs with early stopping of patience 3 based on highest PSNR score. The best training checkpoints from each model were selected, and underwent further training with generator learning rate = 8e-5 and discriminator learning rate = 5e-5 to encourage the learning of subtle anatomical details. Each model was trained for a further 15 epochs with patience of 5. All initial model versions were trained and tested on the 15-slice dataset, while the best performing model was trained and tested on the 100-slice per patient dataset. Fidelity scores of varying configurations are shown in Table 12.

Table 12: SynDiff fidelity comparison across input conditioning strategies.

Input Condition	α	Training Slices	FID	KID	SSIM	PSNR	GMSD	LPIPS
FLAIR	-	15	32.33	0.0205	0.9242	24.33	0.1453	0.0888
T1w	-	15	31.71	0.0218	0.9374	24.84	0.1298	0.0756
T1w + $\alpha\cdot$ Seg	0.1	15	28.72	0.0195	0.9372	25.36	0.1298	0.0736
T1w + $\alpha\cdot$ Seg	0.3	15	26.29	0.0176	0.9383	25.51	0.1267	0.0697
T1w + $\alpha\cdot$ Seg	0.3	100	**18.52**	**0.0105**	**0.9415**	**25.90**	**0.1219**	**0.0592**

Note: α controls the intensity of the scaled segmentation mask that was superimposed onto T1w. Training slices refers to the number of axial slices per patient that the model was trained with. All models save the bottom-most model were trained on 15 axial slices per patient.

6.12 *A4: Custom Diffusion*

The custom diffusion model presented by Mirowski and Fabijańska (2026) is constructed from a U-Net architecture which progresses through alternating downsampling and attention blocks (encoder), followed by alternating upsampling and attention blocks (decoder). As a UNet, the model employs skip connections between sequential encoding and decoding blocks, allowing it to pass on detailed feature information that may otherwise be missed through downsampling. The addition of attention layers helps to preserve global context, allowing the model to retain anatomical information through successive down/upsampling. A example U-Net is shown in Figure 18.

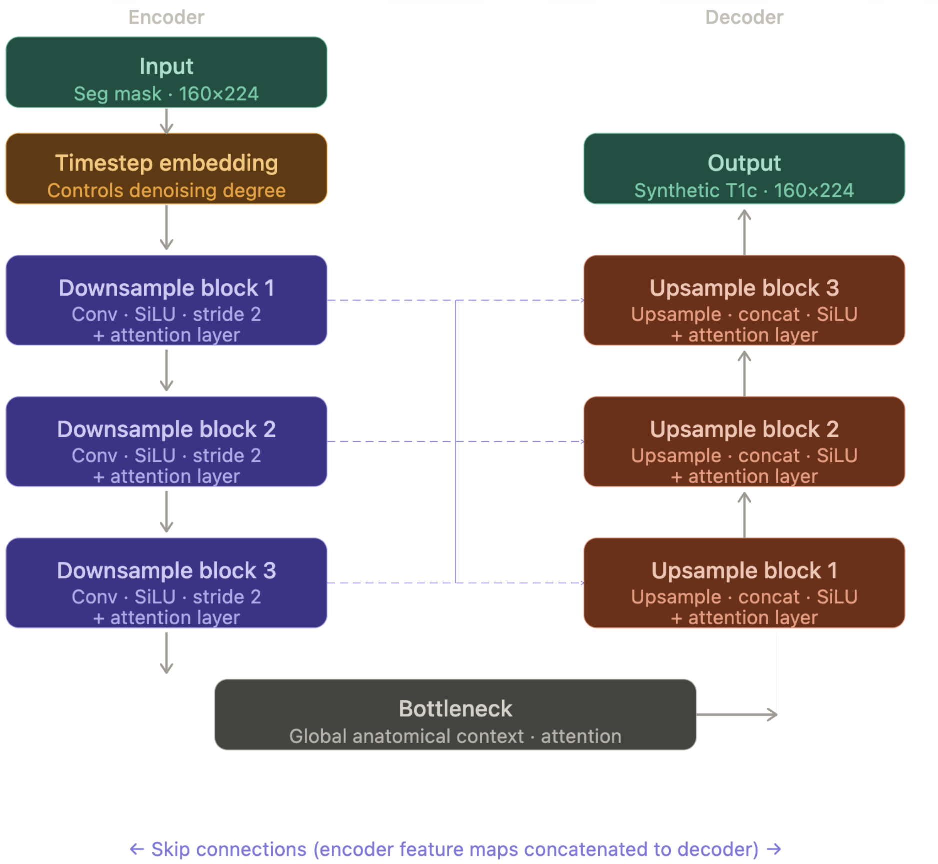

Figure 18: A representation of the U-Net employed in the custom diffusion architecture.

As with Syndiff, custom diffusion utilizes trimestep conditioning so that the model can apply the correct degree of denoising depending on timestep. Interestingly, in place of ReLU, custom diffusion uses sigmoid linear (SiLU) activation, which allows for a slight negative gradient and a smoother gradient profile without risking vanishing gradients. Finally, a dropout rate of 0.3 was used to prevent overfitting—especially important given the limited amount of medical training data. Additional model information can be viewed in Table 13, with progressive model output samples visible in Figure 19.

Table 13: Custom diffusion hyperparameter configurations across training stages.

Parameter	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7	Exp. 8
Corresponding Image	–	a	b	c	d	f	g	h
Conditioned Input	seg	seg	FLAIR	seg	seg	seg	seg	seg
T1c Normalization	[0,1]	[0,1]	[0,1]	[0,1]	[0,1]	[-1,1]	[-1,1]	[-1,1]
LR	1e-4	1e-4	1e-4	1e-4	1e-5	1e-5	1e-5	1e-4
Epochs	200	–	–	–	100	175	200	200
LR Schedule (Linear)	✓	Cosine	✓	✓	✓	✓	✓	✓
Noise Loss (MSE)	✓	✓	✓	SSIM	✓	✓	✓	✓
Intensity Loss	–	–	–	–	0.02	0.02	–	–
Clamping Loss	–	–	–	–	0.04	0.04	–	–
Clamping Range	–	–	–	–	–	[-2,2]	–	–
Assessment	Somewhat correct anatomy; significant black regions; exaggerated brightness	Mode collapse: undifferentiated grey brain regions	Mode collapse: inability to differentiate brain tissue from rest of scan	Mode collapse: noise, with learning of brain boundary	More correct intensity range and sharpness, though incorrect anatomy learning	Mode collapse: lack of anatomical differentiation, though improved intensity	More stable training; insufficient anatomy learning	Mode collapse: noise with minimal tumor region learning

Note: LR = Learning Rate; LR schedule was linear unless otherwise specified; LR warmup was kept at 500 training steps throughout; Noise loss was MSE unless otherwise specified; seg = modified segmentation mask. Epochs less than 100 were not specified, given early model failure. Letters a-g corresond to the output images of Figure 19

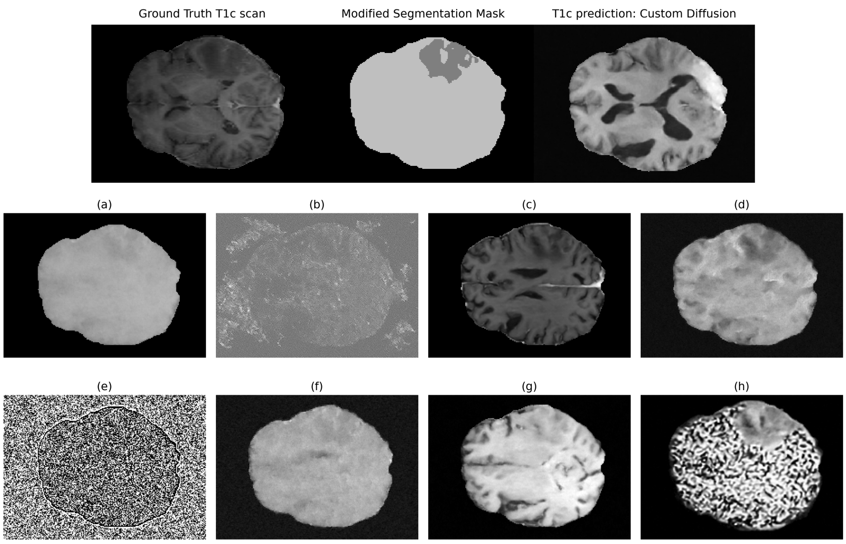

Figure 19: Examples of model outputs at various stages of training. The top right of the figure is the prediction of the best performing generator. The image on the top left is the ground truth T1c. The top-center image is the conditioned modified segmentation mask. Images labeled (a)through (h) represent predicted images following various input configurations and hyperparameter settings. Detailed training details are available in Appendix A4: Custom Diffusion, as well as descriptions of images a-g (Table 13).

APPENDIX B

6.13 *Error Analysis*

Subsequent analysis was conducted at the tumor label level (NCR, Edema, ET) rather than the merged subregion level (TC, WT, ET). The cross-validated BraTS training set is referred to as CV in text, and "training" in figures and tables.

FP and FN rates FN, FP, and complete-miss (Dice = 0) rates were higher in the BraTS test set compared to the other datasets (Figure 20). This was observed across all tumor regions, but was most pronounced in small tumors. For small NCR regions, the BraTS test had an FN rate of 0.514 with a complete miss rate of 18.8%, much higher than for CV (0.250 and 6.7%) and for UCSF (0.149 and 3.5%). Although there were no complete misses for medium NCR, the FN and FP rate for BraTS test (0.234, 0.223) were substantially elevated compared to CV (FN: 0.085, FP: 0.056) and UCSF (FN: 0.053, FP: 0.041). Large NCR had a 1.1% complete miss rate for the BraTS test set, and a much higher FN rate (0.205) than for CV (0.054) and UCSF (0.086). FP rates were slightly more elevated for the BraTS test (0.080) than for UCSF and BraTS CV, both with FP <0.05 (see Table 14).

Table 14: False negative (FN), false positive (FP), and complete miss rate by tumor subregion, size, and dataset. Complete miss is the percentage of cases where the predicted mask has zero overlap with the ground truth. *n* denotes the number of patients. Test refers to the BraTS test set, while train refers to the BraTS training set.

Subregion	Size	Dataset	*n*	FN	FP	Complete miss (%)
NCR	Small	Train	314	0.250	0.110	6.7
		Test	32	0.514	0.351	18.8
		UCSF	172	0.149	0.096	3.5
	Medium	Train	313	0.085	0.056	0.0
		Test	18	0.234	0.223	0.0
		UCSF	134	0.053	0.041	0.0
	Large	Train	323	0.054	0.042	0.0
		Test	189	0.205	0.080	1.1
		UCSF	80	0.086	0.039	1.2

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Table 14 (continued)

Subregion	Size	Dataset	n	FN	FP	Complete miss (%)
Edema	Small	Train	330	0.135	0.098	0.3
		Test	7	0.315	0.332	0.0
		UCSF	264	0.142	0.102	0.0
	Medium	Train	329	0.062	0.056	0.0
		Test	19	0.327	0.281	0.0
		UCSF	182	0.073	0.054	0.0
	Large	Train	340	0.043	0.039	0.0
		Test	225	0.148	0.124	0.0
		UCSF	53	0.069	0.046	0.0
ET	Small	Train	321	0.125	0.082	1.9
		Test	19	0.384	0.220	5.3
		UCSF	220	0.126	0.098	1.8
	Medium	Train	320	0.055	0.051	0.0
		Test	27	0.228	0.176	3.7
		UCSF	165	0.078	0.055	0.0
	Large	Train	330	0.043	0.042	0.0
		Test	196	0.103	0.081	0.0
		UCSF	37	0.079	0.061	0.0

For Edema, while there were no complete misses (save for 0.3% for CV), FN rates in the BraTS test dataset were much higher across all tumor sizes compared to BraTS CV and UCSF: small = 0.315 (CV: 0.135, UCSF: 0.142); medium = 0.327 (CV: 0.062, UCSF: 0.073); large = 0.148 (CV: 0.043, UCSF: 0.069). FP rates between datasets and tumor sizes followed a similar pattern.

Finally, small ET regions in the BraTS test set had a complete miss rate of 5.3% and FN = 0.384, compared to 1.9% and FN = 0.125 in CV, and 1.8% and FN = 0.126. Medium ET again showed elevated FN in the test set (0.228) compared to CV (0.055) and UCSF (0.078), with a miss rate of 3.7% compared to no misses across CV and UCSF. Large ET showed no complete misses across datasets, but still had BraTS with an elevated FN rate (0.103) compared to CV (0.043) and UCSF (0.079).

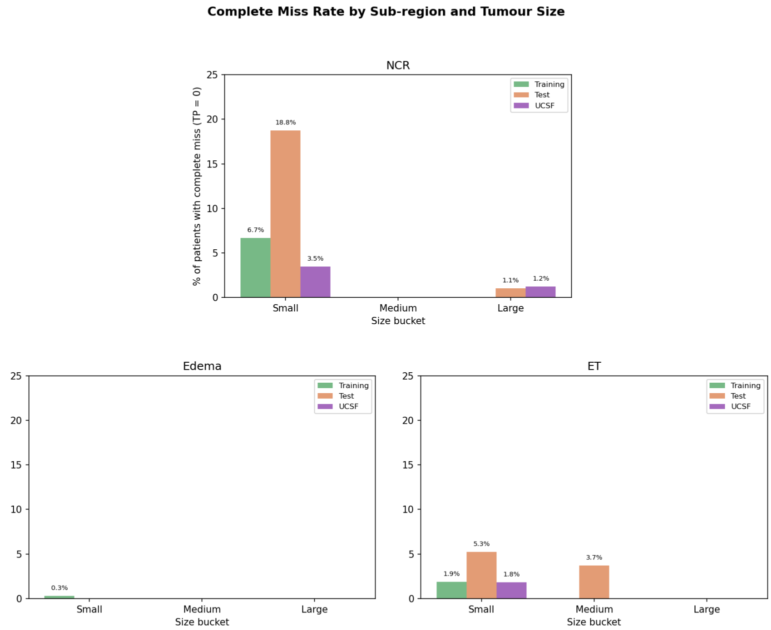

Figure 20: Complete miss rate (TP = 0) between all datasets, tumor subregions, and tumor sizes.

Across all subregions and tumor sizes, the FP and FN rates of UCSF remained consistently close to those of CV, while scores for BraTS test were notably more elevated, especially in small NCR.

Tumor size The BraTS test set contained considerably fewer small and medium tumor cases per region than either the CV or UCSF datasets. For instance, the NCR region of the test set contained 13.4% small, 7.5% large, and 79.1% large tumors – whereas each region was roughly equally represented in the CV dataset. UCSF had more small tumors than CV – for example in edema, with 52.9% small, 36.5% medium, and 10.6% large, again compared to roughly equal representation in CV (Figure 21).

Image Quality Statistics The distributions of both SNR and CNR were highly similar between the CV, test, and UCSF datasets (Figures 22 and 23).

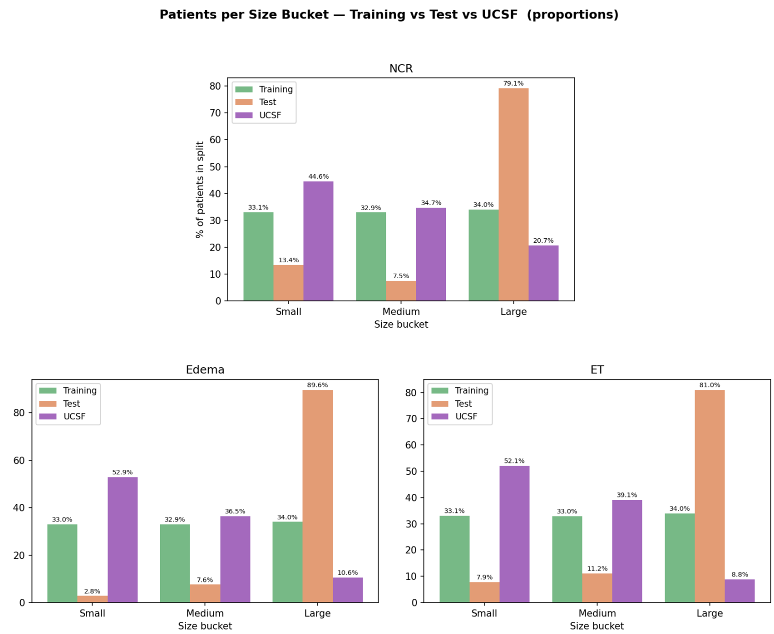

Figure 21: The proportion of large, medium, and small tumors per subregion, per dataset. Tumor size was determined through binning by quantiles, with reference to tumors present in the BraTS training data.

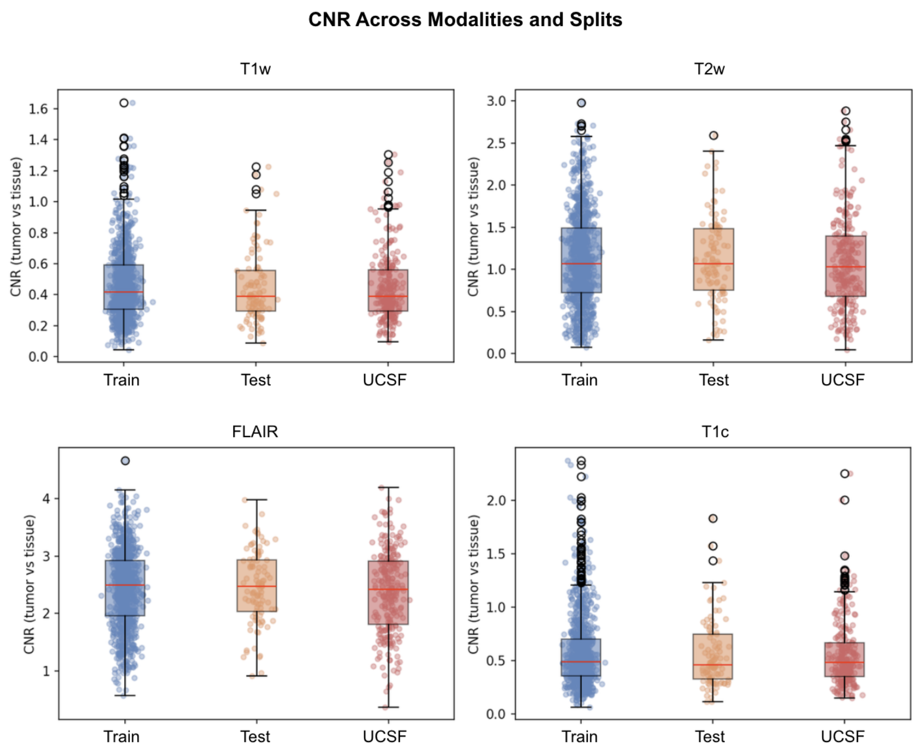

Figure 22: CNR values per scan between datasets.

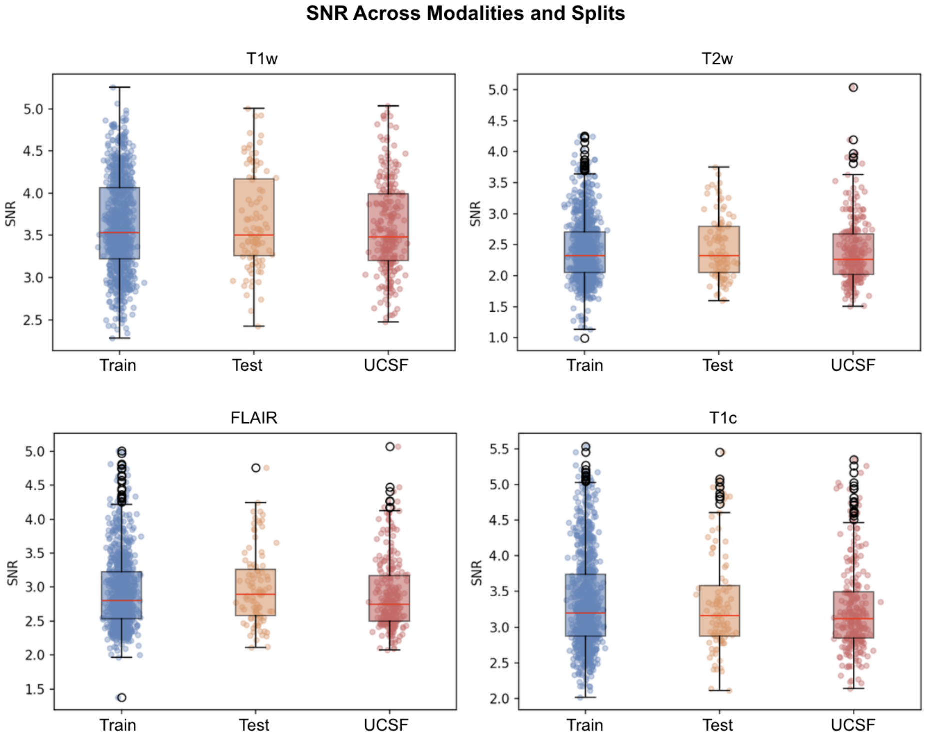

Figure 23: SNR values per scan between datasets.

However, cases of segmentation failure (i.e., individual patients with Dice < 0.3) were associated with significantly increased CNR and SNR in T2w and FLAIR scans within datasets. Table 15 represents an exhaustive list of the SNR and CNR comparisons for individual scans between failed and passing segmentation cases. Effect sizes are reported (W1), as well as significance values (Mann-Whitney p-values).

Table 15: Image quality metrics (SNR and CNR) for segmentation failures (Dice < 0.3) vs. passing cases (Dice ≥ 0.3) across all datasets and tumor subregions. Values are mean $\pm$ standard deviation. W_1 denotes the Wasserstein distance between the failed and passing distributions. p-values are from two-tailed Mann–Whitney U tests; significant results ($p < 0.05$) are shown in bold. There were no failed cases for mean Dice scores in the BraTS training set.

Subregion	Modality	Metric	Failed (mean $\pm$ sd)	Passing (mean $\pm$ sd)	W_1	p
BraTS Training Set	(974 patients, train + val)					
	T1W	SNR	3.588 ± 0.503	3.629 ± 0.552	0.090	0.7438

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NCR

Table 15 (continued)

Subregion	Modality	Type	Failed (mean ± sd)	Passing (mean ± sd)	W_1	p
	T2W	SNR	2.456 ± 0.451	2.415 ± 0.499	0.082	0.5310
	FLAIR	SNR	3.006 ± 0.578	2.919 ± 0.515	0.136	0.1760
	T1C	SNR	3.611 ± 0.789	3.330 ± 0.645	0.281	**0.0456**
	T1W	CNR	0.542 ± 0.304	0.459 ± 0.222	0.111	0.1315
	T2W	CNR	0.950 ± 0.608	1.145 ± 0.551	0.225	**0.0123**
	FLAIR	CNR	2.357 ± 0.587	2.447 ± 0.677	0.140	0.3423
	T1C	CNR	0.588 ± 0.410	0.573 ± 0.321	0.084	0.6948
Edema	T1W	SNR	3.457 ± 0.465	3.623 ± 0.551	0.296	0.6731
	T2W	SNR	2.450 ± 0.341	2.417 ± 0.502	0.239	0.8271
	FLAIR	SNR	2.679 ± 0.323	2.928 ± 0.520	0.294	0.5334
	T1C	SNR	3.264 ± 0.192	3.347 ± 0.663	0.376	0.8235
	T1W	CNR	0.336 ± 0.075	0.464 ± 0.227	0.150	0.4374
	T2W	CNR	1.013 ± 0.789	1.128 ± 0.555	0.430	0.8126
	FLAIR	CNR	1.996 ± 0.542	2.437 ± 0.675	0.488	0.3960
	T1C	CNR	1.064 ± 0.617	0.575 ± 0.329	0.499	0.3344
ET	T1W	SNR	3.665 ± 0.754	3.624 ± 0.547	0.224	0.8003
	T2W	SNR	2.710 ± 0.445	2.406 ± 0.497	0.312	**0.0047**
	FLAIR	SNR	3.432 ± 0.811	2.914 ± 0.508	0.518	**0.0088**
	T1C	SNR	3.590 ± 0.941	3.342 ± 0.655	0.360	0.3757
	T1W	CNR	0.530 ± 0.214	0.463 ± 0.227	0.080	0.1687
	T2W	CNR	1.515 ± 0.533	1.118 ± 0.553	0.400	**0.0035**
	FLAIR	CNR	2.157 ± 0.750	2.447 ± 0.670	0.322	0.0583
	T1C	CNR	0.612 ± 0.276	0.575 ± 0.328	0.077	0.4141
BraTS Test Set (251 patients)						
mean	T1W	SNR	4.060 ± 0.542	3.586 ± 0.541	0.480	0.1771
	T2W	SNR	2.905 ± 0.321	2.385 ± 0.515	0.565	**0.0432**
	FLAIR	SNR	3.713 ± 0.497	2.876 ± 0.515	0.842	**0.0133**
	T1C	SNR	3.247 ± 0.088	3.266 ± 0.634	0.409	0.5706
	T1W	CNR	0.615 ± 0.125	0.444 ± 0.218	0.215	0.0899
	T2W	CNR	1.015 ± 0.609	1.089 ± 0.561	0.293	0.6912

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Table 15 (continued)

Subregion	Modality	Type	Failed (mean ± sd)	Passing (mean ± sd)	W_1	p
	FLAIR	CNR	1.810 ± 0.572	2.388 ± 0.718	0.596	0.1961
	T1C	CNR	0.408 ± 0.154	0.569 ± 0.315	0.179	0.3684
NCR	T1W	SNR	3.764 ± 0.593	3.561 ± 0.528	0.206	0.0916
	T2W	SNR	2.562 ± 0.530	2.369 ± 0.514	0.201	**0.0328**
	FLAIR	SNR	3.221 ± 0.649	2.837 ± 0.486	0.384	**0.0004**
	T1C	SNR	3.409 ± 0.650	3.251 ± 0.632	0.161	0.1145
	T1W	CNR	0.508 ± 0.270	0.437 ± 0.206	0.075	0.2296
	T2W	CNR	1.078 ± 0.618	1.102 ± 0.550	0.129	0.4561
	FLAIR	CNR	2.173 ± 0.889	2.419 ± 0.686	0.341	0.0611
	T1C	CNR	0.582 ± 0.346	0.563 ± 0.309	0.048	0.8527
Edema	T1W	SNR	3.308 ± 0.336	3.597 ± 0.545	0.317	0.2195
	T2W	SNR	2.510 ± 0.417	2.388 ± 0.517	0.207	0.4723
	FLAIR	SNR	2.982 ± 0.353	2.884 ± 0.525	0.233	0.3932
	T1C	SNR	3.053 ± 0.187	3.270 ± 0.635	0.339	0.6264
	T1W	CNR	0.717 ± 0.198	0.441 ± 0.214	0.285	**0.0078**
	T2W	CNR	0.850 ± 0.551	1.093 ± 0.561	0.289	0.2544
	FLAIR	CNR	1.572 ± 0.695	2.397 ± 0.711	0.825	**0.0259**
	T1C	CNR	0.837 ± 0.747	0.562 ± 0.296	0.343	0.9735
ET	T1W	SNR	3.674 ± 0.693	3.582 ± 0.532	0.206	0.9726
	T2W	SNR	2.961 ± 0.602	2.356 ± 0.493	0.625	**0.0010**
	FLAIR	SNR	3.427 ± 0.476	2.854 ± 0.503	0.589	**0.0008**
	T1C	SNR	3.565 ± 0.672	3.251 ± 0.628	0.331	0.1107
	T1W	CNR	0.539 ± 0.222	0.439 ± 0.216	0.111	0.1291
	T2W	CNR	1.303 ± 0.568	1.079 ± 0.558	0.239	0.1689
	FLAIR	CNR	2.037 ± 0.609	2.401 ± 0.720	0.377	0.0903
	T1C	CNR	0.571 ± 0.326	0.565 ± 0.312	0.094	0.9554
UCSF Dataset (501 patients)						
	T1W	SNR	3.484 ± 0.564	3.438 ± 0.562	0.134	0.9125
	T2W	SNR	2.114 ± 0.272	2.056 ± 0.275	0.104	0.6090

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mean

Table 15 (continued)

Subregion	Modality	Type	Failed (mean ± sd)	Passing (mean ± sd)	W_1	p
	FLAIR	SNR	2.558 ± 0.115	2.514 ± 0.190	0.093	0.4247
	T1C	SNR	3.118 ± 0.104	2.925 ± 0.311	0.237	0.0896
	T1W	CNR	0.438 ± 0.428	0.445 ± 0.265	0.174	0.3536
	T2W	CNR	0.935 ± 0.687	0.852 ± 0.506	0.266	0.9341
	FLAIR	CNR	1.574 ± 0.787	2.050 ± 0.583	0.481	0.2627
	T1C	CNR	1.216 ± 0.669	0.538 ± 0.298	0.677	**0.0029**
NCR	T1W	SNR	3.490 ± 0.636	3.494 ± 0.572	0.120	0.8625
	T2W	SNR	2.137 ± 0.199	2.032 ± 0.272	0.114	**0.0419**
	FLAIR	SNR	2.532 ± 0.150	2.499 ± 0.193	0.056	0.2013
	T1C	SNR	2.960 ± 0.285	2.894 ± 0.322	0.083	0.2519
	T1W	CNR	0.551 ± 0.353	0.407 ± 0.225	0.145	0.0672
	T2W	CNR	1.257 ± 0.629	0.800 ± 0.473	0.457	**0.0002**
	FLAIR	CNR	2.174 ± 0.562	2.027 ± 0.598	0.158	0.2497
	T1C	CNR	0.672 ± 0.341	0.495 ± 0.245	0.177	**0.0018**
Edema	T1W	SNR	3.268 ± 0.490	3.441 ± 0.562	0.230	0.4657
	T2W	SNR	2.004 ± 0.229	2.057 ± 0.276	0.125	0.8533
	FLAIR	SNR	2.451 ± 0.134	2.515 ± 0.189	0.102	0.5358
	T1C	SNR	2.995 ± 0.176	2.927 ± 0.311	0.149	0.6878
	T1W	CNR	0.250 ± 0.123	0.447 ± 0.267	0.197	0.1561
	T2W	CNR	0.474 ± 0.322	0.856 ± 0.509	0.391	0.1573
	FLAIR	CNR	1.517 ± 0.413	2.052 ± 0.582	0.547	0.1299
	T1C	CNR	0.816 ± 0.629	0.540 ± 0.296	0.332	0.7162
ET	T1W	SNR	3.158 ± 0.267	3.485 ± 0.570	0.349	**0.0300**
	T2W	SNR	2.183 ± 0.233	2.037 ± 0.273	0.148	**0.0117**
	FLAIR	SNR	2.526 ± 0.105	2.502 ± 0.190	0.073	0.2670
	T1C	SNR	3.055 ± 0.244	2.906 ± 0.319	0.162	**0.0093**
	T1W	CNR	0.590 ± 0.298	0.413 ± 0.229	0.183	**0.0017**
	T2W	CNR	1.278 ± 0.505	0.803 ± 0.482	0.479	**<0.0001**
	FLAIR	CNR	2.263 ± 0.605	2.025 ± 0.601	0.316	**0.0238**
	T1C	CNR	0.782 ± 0.482	0.497 ± 0.247	0.291	**0.0011**